INCH POUND

MIL-M-38510/79D
17 AUGUST 2005
SUPERSEDING
MIL-M-38510/79C
5 NOVEMBER 1987

MILITARY SPECIFICATION

# MICROCIRCUITS, DIGITAL, BIPOLAR, SCHOTTKY TTL, DATA SELECTORS / MULTIPLEXERS WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

Inactive for new design after 23 August 1996.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF 38535.

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Schottky TTL, data selectors / multiplexers (three-state) microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.4).

1.2 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device type. The device type is as follows:

| Device type | Circuit |
|---|---|
| 01 | 8 input, data selector / multiplexer |
| 02 | Dual, 4-input, data selector / mutiplexer |
| 03 | Quad, 2-input, data selector / multiplexer |
| 04 | Quad, 2-input, data selector / multiplexer with inverted output |
| 05 | 8-input, data selector / mutiplexer with 3-state outputs |
| 06 | Quad, 2-input, data selector / multiplexer with 3-state outputs |
| 07 | Quad, 2-input, data selector / multiplexer with 3-state inverted output |
| 08 | Dual, 4-input, data selector / multiplexer with 3-state outputs |

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

| Outline letter | Descriptive designator | Terminals | Package style |
|---|---|---|---|
| E | GDIP1-T16, CDIP2-T16 | 16 | Dual in line package |
| F | GDFP2-F16, CDFP3-F16 | 16 | Flat Package |
| 2 | CQCC1-N20 | 20 | Square chip |
| X | CQCC2-N20 | 20 | Square chip carrier package |

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, P. O. Box 3990, Columbus, OH 43218-3990, or emailed to bipolar@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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1.3 Absolute maximum ratings.

| | |
|---|---|
| Supply voltage range | -0.5 V dc to +7.0 V dc |
| Input voltage range | -1.2 V dc at -18 mA to +5.5 V dc |
| Storage temperature range | -65°C to +150°C |
| Maximum power dissipation ($P_D$) 1/ : | |
| Device types 01 and 02 | 385 mW |
| Device type 03 | 430 mW |
| Device type 04 | 336 mW |
| Device type 05 | 468 mW |
| Device type 06 | 545 mW |
| Device type 07 | 479 mW |
| Device type 08 | 550 mW |
| Lead temperature (soldering 10 seconds) | 300°C |
| Thermal resistance, junction-to-case ($\theta_{JC}$) : | |
| Cases E, F, 2, and X | (See MIL-STD-1835) |
| Junction temperature ($T_J$) 2/ | +175°C |

1.4 Recommended operating conditions.

| | |
|---|---|
| Supply voltage ($V_{CC}$) | 4.5 V dc minimum to 5.5 V dc maximum |
| Minimum high level input voltage ($V_{IH}$) | 2.0 V dc |
| Maximum low level input voltage ($V_{IL}$) | 0.8 V dc 3/ |
| Case operating temperature range ($T_C$) | -55°C to 125°C |

2.0 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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1/ Must withstand the added $P_D$ due to short circuit condition (e.g., $I_{OS}$) test.
2/ Maximum junction temperature should not be exceeded except in accordance with allowable short duration burn-in screening conditions in accordance with MIL-PRF-38535.
3/ $V_{IL}$ = 0.7 V at +125 ° C



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3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.3).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.3.2 Truth table. The truth table shall be as specified on figure 2.

3.3.3 Schematic circuits. The schematic circuit shall be maintained by the manufacturer and made available to the qualifying activity and the preparing activity upon request.

3.3.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in Table II. The electrical tests for each subgroup are described in Table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535 and 1.2 herein.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-PRF-38535, appendix A).



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TABLE I. Electrical performance characteristics.

| Test | Symbol | Conditions<br>$-55°C \le T_C \le +125°C$<br>Unless otherwise specified | | Device type | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|---|
| High level output voltage | $V_{OH}$ | $V_{CC}$ = 4.5 V; $V_{IL}$ = 0.8 V<br>$I_{OH}$ = -1.0 mA<br>@$T_C$ = 125°C, $V_{IL}$ = 0.7 V | | 01, 02<br>03, 04 | 2.5 | | V |
| | | $V_{CC}$ = 4.5 V; $V_{IL}$ = 0.8 V<br>$I_{OH}$ = -2.0 mA<br>@$T_C$ = 125°C, $V_{IL}$ = 0.7 V | | 05, 06 | 2.4 | | |
| | | $V_{CC}$ = 4.5 V; $V_{IL}$ = 0.8 V<br>$I_{OH}$ = -1.0 mA<br>@$T_C$ = 125°C, $V_{IL}$ = 0.7 V | | 07, 08 | | | |
| Low level output voltage | $V_{OL}$ | $V_{CC}$ = 4.5 V<br>$I_{OL}$ = 20 mA | | ALL | | .5 | V |
| | | $T_C$ = 125°C | | ALL | | .45 | V |
| Input clamp voltage | $V_{IC}$ | $V_{CC}$ = 4.5 V<br>$I_{IN}$ = -18 mA, $T_C$ = 25°C | | ALL | | -1.2 | V |
| Off state output current | $I_{OFF1}$ | $V_{CC}$ = 5.5 V<br>$V_O$ = 2.7 V | | 05, 06<br>07, 08 | | 50 | µA |
| Off state output current | $I_{OFF2}$ | $V_{CC}$ = 5.5 V<br>$V_O$ = 0.5 V | | 05, 06<br>07, 08 | | -50 | µA |
| High level input current | $I_{IH1}$ | $V_{CC}$ = 5.5 V<br>$V_{IN}$ = 2.7 V | All inputs | 01, 02<br>05, 08 | | 50 | µA |
| | | | A and B inputs | 03, 04 | | | |
| | | | All inputs except S | 06, 07 | | | |
| | | | S and G inputs | 03, 04 | | 100 | |
| | | | S input | 06, 07 | | | |
| High level input current | $I_{IH2}$ | $V_{CC}$ = 5.5 V<br>$V_{IN}$ = 5.5 V | All inputs | All | | 1.0 | mA |
| Low level input current | $I_{IL}$ | $V_{CC}$ = 5.5 V<br>$V_{IN}$ = 0.5 V | All inputs | 01, 02<br>05, 08 | -1.0 | -2.0 | mA |
| | | | A and B inputs | 03, 04 | 0.1 | | |
| | | | All inputs except S | 06, 07 | -1.0 2/ | | |
| | | | S and G inputs | 03, 04 | 0.1 | -4.0 | |
| | | | S input | 06, 07 | -2.0 2/ | -4.0 | |

See footnotes at end of table.



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TABLE I. Electrical performance characteristics - Continued.

| Test | Symbol | Conditions<br>$-55°C \le T_C \le +125°C$<br>Unless otherwise specified | Device type | Limits | | Units |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Short circuit output current | $I_{OS}$ | $V_{CC}$ = 5.5 V 1/` | All | -40 | -110 | mA |
| Supply current | $I_{CC}$ | $V_{CC}$ = 5.5 V | 01, 02 | | 70 | mA |
| | | | 03 | | 78 | |
| | | | 04 | | 61 | |
| Supply current | $I_{CCO}$ | $V_{CC}$ = 5.5 V<br>$V_{IN}$ = 5.5 V | 05 | | 85 | mA |
| | | | 06 | | 99 | |
| | | | 07 | | 87 | |
| | | | 08 | | 100 | |
| Supply current | $I_{CC1}$ | $V_{CC}$ = 5.5 V | 08 | | 80 | mA |
| Collector cut-off current | $I_{CEX}$ | $V_{CC}$ = 5.5 V, $V_{OH}$ = 5.5 V<br>$V_{IL}$ = GND, $V_{IH}$ = 5.5 V | 01 thru 08 | | 250 | µA |
| Low level supply current | $I_{CCL}$ | $V_{CC}$ = 5.5 V | 06 | | 93 | mA |
| | | | 07 | | 81 | |
| High level supply current | $I_{CCH}$ | $V_{CC}$ = 5.5 V | 06 | | 68 | mA |
| | | | 07 | | 56 | |
| From A, B, C, to Y | $t_{PLH2}$ | $V_{CC}$ = 5.0 V<br>$C_L$ = 50 pF<br>$R_L$ = 280Ω | 01 | 2.0 | 26.0 | ns |
| From A, B, C, to Y | $t_{PHL2}$ | | 01 | 2.0 | 26.0 | ns |
| From A, B, C, to W | $t_{PLH1}$ | | 01 | 2.0 | 22.0 | ns |
| From A, B, C, to W | $t_{PHL1}$ | | 01 | 2.0 | 20.0 | ns |
| From any D to Y | $t_{PLH6}$ | | 01 | 2.0 | 18.0 | ns |
| From any D to Y | $t_{PHL6}$ | | 01 | 2.0 | 18.0 | ns |
| From any D to W | $t_{PLH5}$ | | 01 | 2.0 | 11.5 | ns |

See footnotes at end of table.



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TABLE I. Electrical performance characteristics - Continued.

| Test | Symbol | Conditions<br>$-55°C \le T_C \le +125°C$<br>Unless otherwise specified | Device type | Limits | | Units |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| From any D to W | $t_{PHL5}$ | $V_{CC}$ = 5.0 V<br>$C_L$ = 50 pF<br>$R_L$ = 280 Ω | 01 | 2.0 | 11.5 | ns |
| From strobe to Y | $t_{PLH4}$ | | 01 | 2.0 | 24.0 | ns |
| From strobe to Y | $t_{PHL4}$ | | 01 | 2.0 | 26.0 | ns |
| From strobe to W | $t_{PLH3}$ | | 01 | 2.0 | 19.5 | ns |
| From strobe to W | $t_{PHL3}$ | | 01 | 2.0 | 18.0 | ns |
| From data to Y | $t_{PLH1}$ | | 02, 08 | 2.0 | 14.5 | ns |
| From data to Y | $t_{PHL1}$ | | 02, 08 | 2.0 | 14.5 | ns |
| From select to Y | $t_{PLH2}$ | | 02, 08 | 2.0 | 26.0 | ns |
| From select to Y | $t_{PHL2}$ | | 02, 08 | 2.0 | 26.0 | ns |
| From strobe to Y | $t_{PLH3}$ | | 02 | 2.0 | 22.0 | ns |
| From strobe to Y | $t_{PHL3}$ | | 02 | 2.0 | 21.0 | ns |
| From data to Y | $t_{PLH2}$ | | 03 | 2.0 | 12.0 | ns |
| | | | 04 | | 11.0 | |
| From data to Y | $t_{PHL2}$ | | 03 | 2.0 | 12.0 | ns |
| | | | 04 | | 11.0 | |
| From strobe to Y | $t_{PLH3}$ | | 03 | 2.0 | 18.0 | ns |
| | | | 04 | | 18.0 | |
| From strobe to Y | $t_{PHL3}$ | | 03 | 2.0 | 18.5 | ns |
| | | | 04 | | 18.5 | |

See footnotes at end of table.



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TABLE I. Electrical performance characteristics - Continued.

| Test | Symbol | Conditions<br>$-55°C \le T_C \le +125°C$<br>Unless otherwise specified | Device type | Limits | | Units |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| From select to Y | $t_{PLH1}$ | $V_{CC}$ = 5.0 V<br>$C_L$ = 50 pF<br>$R_L$ = 280 Ω | 03 | 2.0 | 18.5 | ns |
| | | | 04 | | 18.5 | |
| From select to Y | $t_{PHL1}$ | | 03 | 2.0 | 18.5 | ns |
| | | | 04 | | 18.5 | |
| From A, B, C to Y | $t_{PLH2}$ | | 05 | 2.0 | 26.0 | ns |
| From A, B, C to Y | $t_{PHL2}$ | | 05 | 2.0 | 28.0 | ns |
| From A, B, C to W | $t_{PLH1}$ | | 05 | 2.0 | 22.0 | ns |
| From A, B, C to W | $t_{PHL1}$ | | 05 | 2.0 | 20.0 | ns |
| From any D to Y | $t_{PLH4}$ | | 05 | 2.0 | 18.0 | ns |
| From any D to Y | $t_{PHL4}$ | | 05 | 2.0 | 18.0 | ns |
| From any D to W | $t_{PLH3}$ | | 05 | 2.0 | 11.5 | ns |
| From any D to W | $t_{PHL3}$ | | 05 | 2.0 | 11.5 | ns |
| From strobe to Y | $t_{ZH3}$ | | 05 | 2.0 | 25.5 | ns |
| From strobe to Y | $t_{ZL3}$ | | 05 | 2.0 | 27.5 | ns |
| From strobe to Y | $t_{HZ4}$ | | 05 | 2.0 | 24.0 | ns |
| From strobe to Y | $t_{LZ4}$ | | 05 | 2.0 | 22.0 | ns |
| From strobe to W | $t_{ZH1}$ | | 05 | 2.0 | 25.5 | ns |
| From strobe to W | $t_{ZL1}$ | | 05 | 2.0 | 27.5 | ns |
| From strobe to W | $t_{HZ2}$ | | 05 | 2.0 | 24.0 | ns |

See footnotes at end of table.



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TABLE I. Electrical performance characteristics - Continued.

| Test | Symbol | Conditions<br>$-55°C \le T_C \le +125°C$<br>Unless otherwise specified | Device type | Limits | | Units |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| From strobe to W | $t_{LZ2}$ | $V_{CC}$ = 5.0 V<br>$C_L$ = 50 pF<br>$R_L$ = 280 Ω | 05 | 2.0 | 22.0 | ns |
| From select to Y | $t_{PLH1}$ | | 06 | 2.0 | 22.0 | ns |
| | | | 07 | | 18.5 | |
| From select to Y | $t_{PHL1}$ | | 06 | 2.0 | 22.0 | ns |
| | | | 07 | | 18.5 | |
| From data to Y | $t_{PLH2}$ | | 06 | 2.0 | 12.0 | ns |
| | | | 07 | | 10.0 | |
| From data to Y | $t_{PHL2}$ | | 06 | 2.0 | 11.0 | ns |
| | | | 07 | | 10.0 | |
| From output control to Y | $t_{ZH}$ | | 05, 06 | 2.0 | 28.0 | ns |
| | | | 07 | | 28.0 | |
| | | | 08 | | 30.0 | |
| From output control to Y | $t_{ZL}$ | | 05, 06 | 2.0 | 30.0 | ns |
| | | | 07 | | 30.0 | |
| | | | 08 | | 31.0 | |
| From output control to Y | $t_{HZ}$ | | 05, 06 | 2.0 | 24.0 | ns |
| | | | 07 | | 24.0 | |
| | | | 08 | | 18.0 | |
| From output control to Y | $t_{LZ}$ | | 05, 06 | 2.0 | 22.0 | ns |
| | | | 07 | | 22.0 | |
| | | | 08 | | 20.0 | |

1/ Not more than one output should be shorted at one time.
2/ For device type 06, $I_{IL}$ minimum limit shall be -0.005 mA for circuit B.



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TABLE II. Electrical test requirements.

| MIL-PRF-38535 Test requirements | Subgroups (see table III) | |
|---|---|---|
| | Class S Devices | Class B Devices |
| Interim electrical parameters | 1 | 1 |
| Final electrical test parameters | 1*, 2, 3, 7, 9, 10, 11 | 1*, 2, 3, 7,9 |
| Group A test requirements | 1, 2, 3, 7, 8, 9, 10, 11 | 1, 2, 3, 7, 8, 9, 10, 11 |
| Group B electrical test parameters when using the method 5005 QCI option | 1, 2, 3, 7, 8, 9, 10, 11 | N/A |
| Group C end-point electrical Parameters | 1, 2, 3, 7, 8, 9, 10, 11 | 1, 2, 3 |
| Group D end point electrical Parameters | 1, 2, 3 | 1, 2, 3 |

*PDA applies to subgroup 1

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.3 Screening. Screening shall be in accordance with MIL-PRF-38535, and shall be conducted on all devices prior to qualification and conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535.



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4.4 Technology Conformance Inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.
b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.



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| Terminal number case | | Terminal name Device type 01 | | Terminal name Device type 02 | | Terminal name Device type 03 | | Terminal name Device type 04 | | Terminal name Device type 05 | | Terminal name Device type 06 | | Terminal name Device type 07 | | Terminal name Deviec type 08 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2, X | E, F | 2, X | E, F | 2, X | E, F | 2, X | E, F | 2, X | E, F | 2, X | E, F | 2, X | E, F | 2, X | E, F | 2, X | E, F |
| 1 | 1 | NC | D3 | NC | 1G | NC | S | NC | S | NC | D3 | NC | S | NC | S | NC | 1G |
| 2 | 2 | D3 | D2 | 1G | B | S | 1A | S | 1A | D3 | D2 | S | 1A | S | 1A | 1G | B |
| 3 | 3 | D2 | D1 | B | 1C3 | 1A | 1B | 1A | 1B | D2 | D1 | 1A | 1B | 1A | 1B | B | 1C3 |
| 4 | 4 | D1 | D0 | 1C3 | 1C2 | 1B | 1Y | 1B | 1Y | D1 | D0 | 1B | 1Y | 1B | 1Y | 1C3 | 1C2 |
| 5 | 5 | D0 | Y | 1C2 | 1C1 | 1Y | 2A | 1Y | 2A | D0 | Y | 1Y | 2A | 1Y | 2A | 1C2 | 1C1 |
| 6 | 6 | NC | W | NC | 1C0 | NC | 2B | NC | 2B | NC | W | NC | 2B | NC | 2B | NC | 1C0 |
| 7 | 7 | Y | ST | 1C1 | 1Y | 2A | 2Y | 2A | 2Y | Y | ST | 2A | 2Y | 2A | 2Y | 1C1 | 1Y |
| 8 | 8 | W | GND | 1C0 | GND | 2B | GND | 2B | GND | W | GND | 2B | GND | 2B | GND | 1C0 | GND |
| 9 | 9 | ST | C | 1Y | 2Y | 2Y | 3Y | 2Y | 3Y | ST | C | 2Y | 3Y | 2Y | 3Y | 1Y | 2Y |
| 10 | 10 | GND | B | GND | 2C0 | GND | 3B | GND | 3B | GND | B | GND | 3B | GND | 3B | GND | 2C0 |
| 11 | 11 | NC | A | NC | 2C1 | NC | 3A | NC | 3A | NC | A | NC | 3A | NC | 3A | NC | 2C1 |
| 12 | 12 | C | D7 | 2Y | 2C2 | 3Y | 4Y | 3Y | 4Y | C | D7 | 3Y | 4Y | 3Y | 4Y | 2Y | 2C2 |
| 13 | 13 | B | D6 | 2C0 | 2C3 | 3B | 4B | 3B | 4B | B | D6 | 3B | 4B | 3B | 4B | 2C0 | 2C3 |
| 14 | 14 | A | D5 | 2C1 | A | 3A | 4A | 3A | 4A | A | D5 | 3A | 4A | 3A | 4A | 2C1 | A |
| 15 | 15 | D7 | D4 | 2C2 | 2G | 4Y | G | 4Y | G | D7 | D4 | 4Y | 0E | 4Y | 0E | 2C2 | 2G |
| 16 | 16 | NC | $V_{CC}$ | NC | $V_{CC}$ | NC | $V_{CC}$ | NC | $V_{CC}$ | NC | $V_{CC}$ | NC | $V_{CC}$ | NC | $V_{CC}$ | NC | $V_{CC}$ |
| 17 | | D6 | | 2C3 | | 4B | | 4B | | D6 | | 4B | | 4B | | 2C3 | |
| 18 | | D5 | | A | | 4A | | 4A | | D5 | | 4A | | 4A | | A | |
| 19 | | D4 | | 2G | | G | | G | | D4 | | 0E | | 0E | | 2G | |
| 20 | | $V_{CC}$ | | $V_{CC}$ | | $V_{CC}$ | | $V_{CC}$ | | $V_{CC}$ | | $V_{CC}$ | | $V_{CC}$ | | $V_{CC}$ | |

FIGURE 1. Terminal connections.



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Device types 01 and 05

| Inputs | | | | | | | | | | | | Outputs | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Select | | | Strobe | Data | | | | | | | | Type 01 | | Type 05 | |
| C | B | A | S | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 | Y | W | Y | W |
| X | X | X | H | X | X | X | X | X | X | X | X | L | H | Z | Z |
| L | L | L | L | L | X | X | X | X | X | X | X | L | H | L | H |
| L | L | L | L | H | X | X | X | X | X | X | X | H | L | H | L |
| L | L | H | L | X | L | X | X | X | X | X | X | L | H | L | H |
| L | L | H | L | X | H | X | X | X | X | X | X | H | L | H | L |
| L | H | L | L | X | X | L | X | X | X | X | X | L | H | L | H |
| L | H | L | L | X | X | H | X | X | X | X | X | H | L | H | L |
| L | H | H | L | X | X | X | L | X | X | X | X | L | H | L | H |
| L | H | H | L | X | X | X | H | X | X | X | X | H | L | H | L |
| H | L | L | L | X | X | X | X | L | X | X | X | L | H | L | H |
| H | L | L | L | X | X | X | X | H | X | X | X | H | L | H | L |
| H | L | H | L | X | X | X | X | X | L | X | X | L | H | L | H |
| H | L | H | L | X | X | X | X | X | H | X | X | H | L | H | L |
| H | H | L | L | X | X | X | X | X | X | L | X | L | H | L | H |
| H | H | L | L | X | X | X | X | X | X | H | X | H | L | H | L |
| H | H | H | L | X | X | X | X | X | X | X | L | L | H | L | H |
| H | H | H | L | X | X | X | X | X | X | X | H | H | L | H | L |

H = high logic level, L = low logic level, X = irrelevant, Z = high impedance

FIGURE 2. Truth tables.



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Device type 02

| Select Inputs | | Data inputs | | | | Strobe | Outputs |
|---|---|---|---|---|---|---|---|
| B | A | C0 | C1 | C2 | C3 | G | Y |
| X | X | X | X | X | X | H | L |
| L | L | L | X | X | X | L | L |
| L | L | H | X | X | X | L | H |
| L | H | X | L | X | X | L | L |
| L | H | X | H | X | X | L | H |
| H | L | X | X | L | X | L | L |
| H | L | X | X | H | X | L | H |
| H | H | X | X | X | L | L | L |
| H | H | X | X | X | H | L | H |

Address inputs A and B are common to both sections.
H = high level, L = low level, X = irrelevant

Device types 03 and 04

| Inputs | | | | Output Y | |
|---|---|---|---|---|---|
| Strobe | Select | A | B | Type 03 | Type 04 |
| H | X | X | X | L | H |
| L | L | L | X | L | H |
| L | L | H | X | H | L |
| L | H | X | L | L | H |
| L | H | X | H | H | L |

H = high level, L = low level, X = irrelevant.

FIGURE 2. Truth tables – Continued.



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Device types 06 and 07

| Inputs | | | | Output Y | |
|---|---|---|---|---|---|
| Output Control | Select | A | B | Type 06 | Type 07 |
| H | X | X | X | Z | Z |
| L | L | L | X | L | H |
| L | L | H | X | H | L |
| L | H | X | L | L | H |
| L | H | X | H | H | L |

H = high level, L = low level, X = irrelevant,
Z = high impedance (off).

Function table

Device type 08

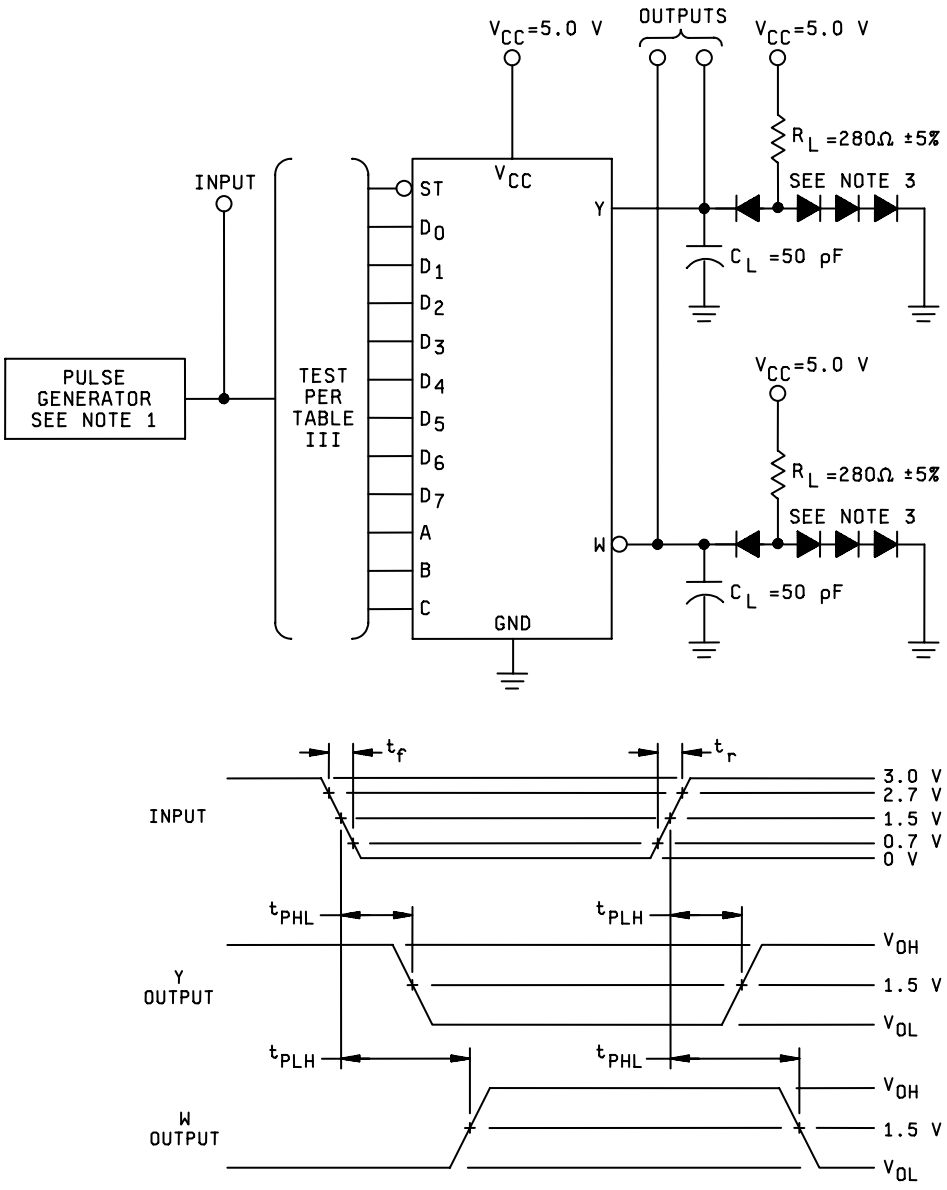
| Select Inputs | | Data inputs | | | | Output control | Output |
|---|---|---|---|---|---|---|---|
| B | A | C0 | C1 | C2 | C3 | G | Y |
| X | X | X | X | X | X | H | Z |
| L | L | L | X | X | X | L | L |
| L | L | H | X | X | X | L | H |
| L | H | X | L | X | X | L | L |
| L | H | X | H | X | X | L | H |
| H | L | X | X | L | X | L | L |
| H | L | X | X | H | X | L | H |
| H | H | X | X | X | L | L | L |
| H | H | X | X | X | H | L | H |

Address inputs A and B are common to both sections.
H = high level, L = low level, X = irrelevant,
Z = high impedance (off).

FIGURE 2. Truth tables - Continued.



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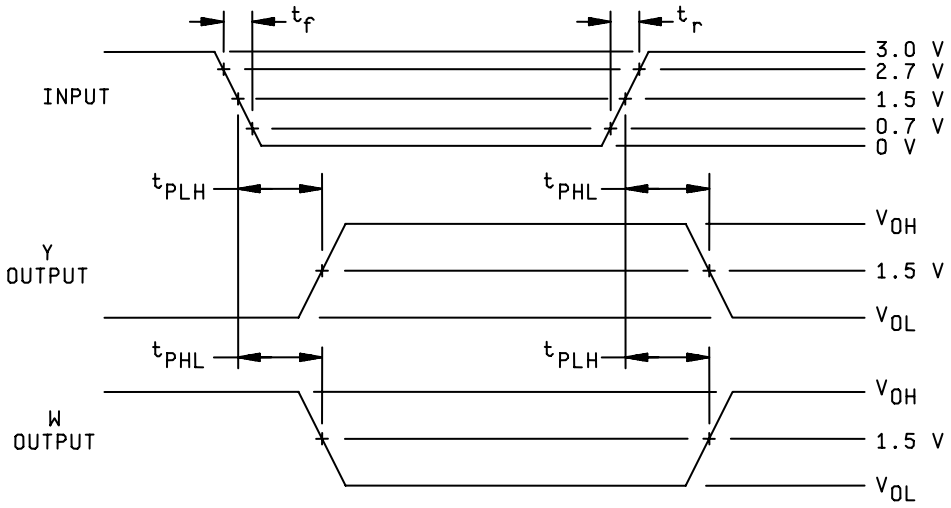

NOTES:

1. The input pulse has the following characteristics: $t_r = t_f \leq 2.5$ ns, PRR $\leq$ 1 MHz, and $Z_{OUT} \approx 50\ \Omega$.
2. $C_L$ includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Only the output under test needs to be loaded.

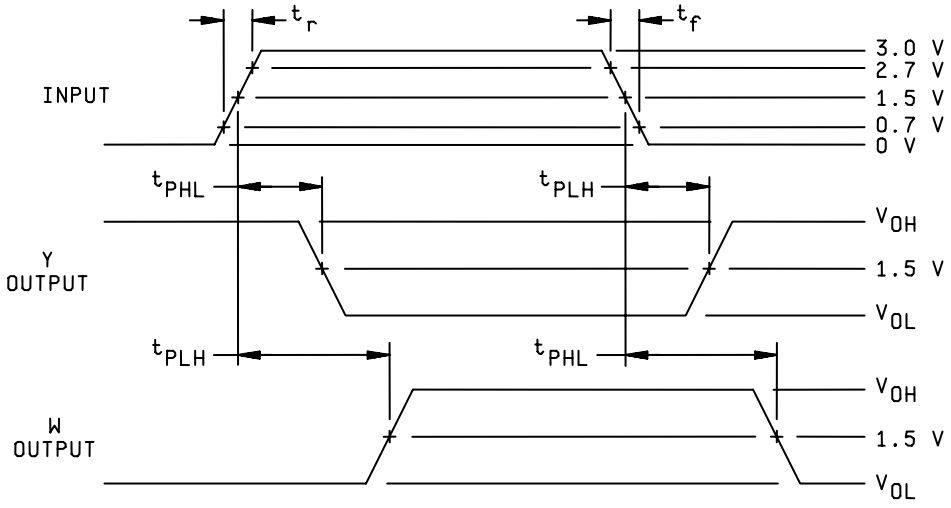
FIGURE 3. Switching time test circuits and waveforms for device type 01.



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STROBE TO OUTPUT VOLTAGE WAVEFORMS - TYPE 01

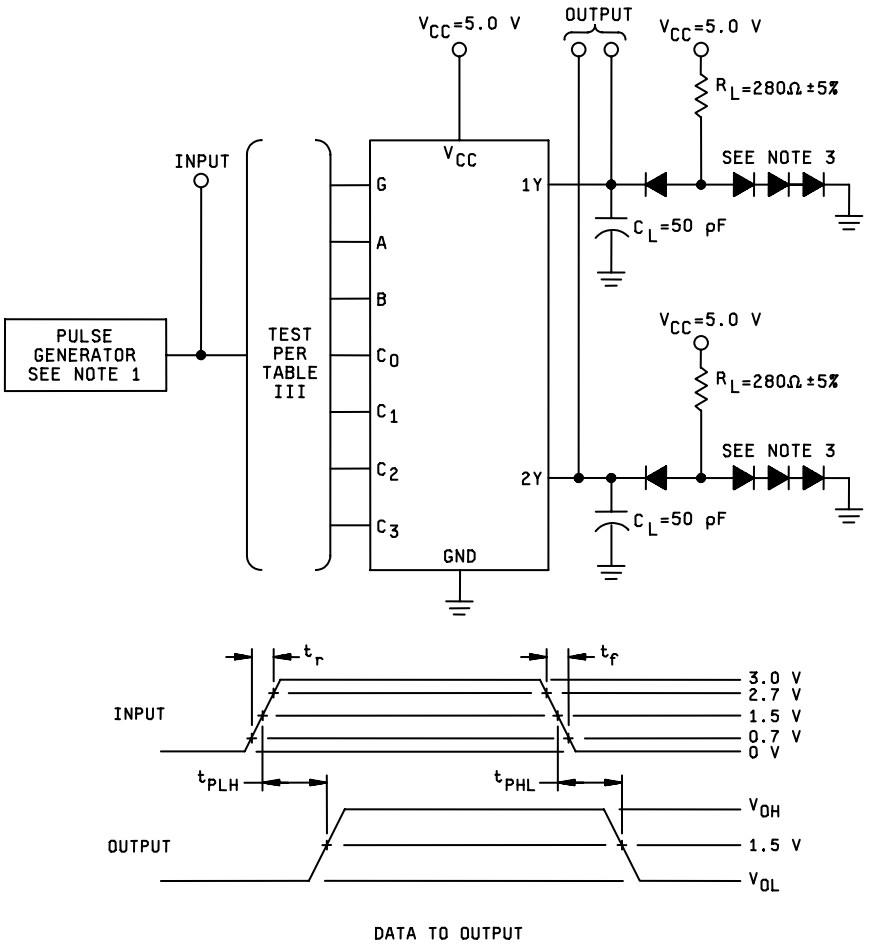



DATA TO OUTPUT VOLTAGE WAVEFORMS - TYPE 01

FIGURE 3. Switching time test circuits and waveforms for device type 01- Continued.



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NOTES:

1. The input pulse has the following characteristics: $t_r = t_f \leq 2.5$ ns, PRR $\leq$ 1 MHz, and $Z_{OUT} \approx 50\ \Omega$.
2. $C_L$ includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Only the output under test needs to be loaded.

FIGURE 4. Switching time test circuits and waveforms for device type 02.



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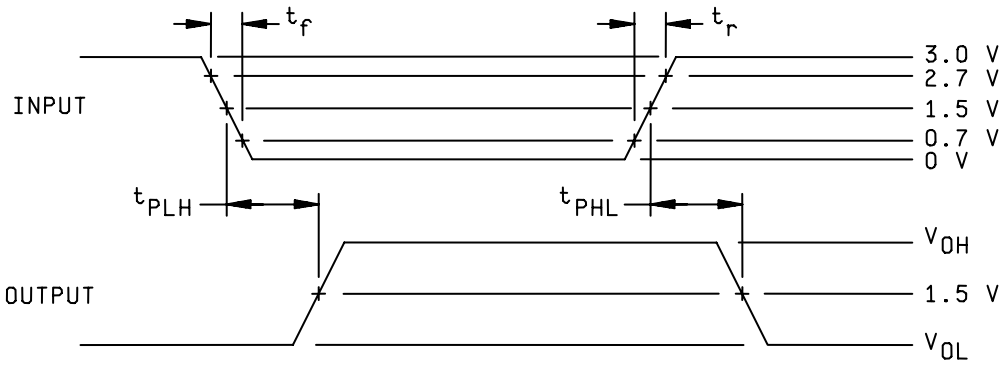
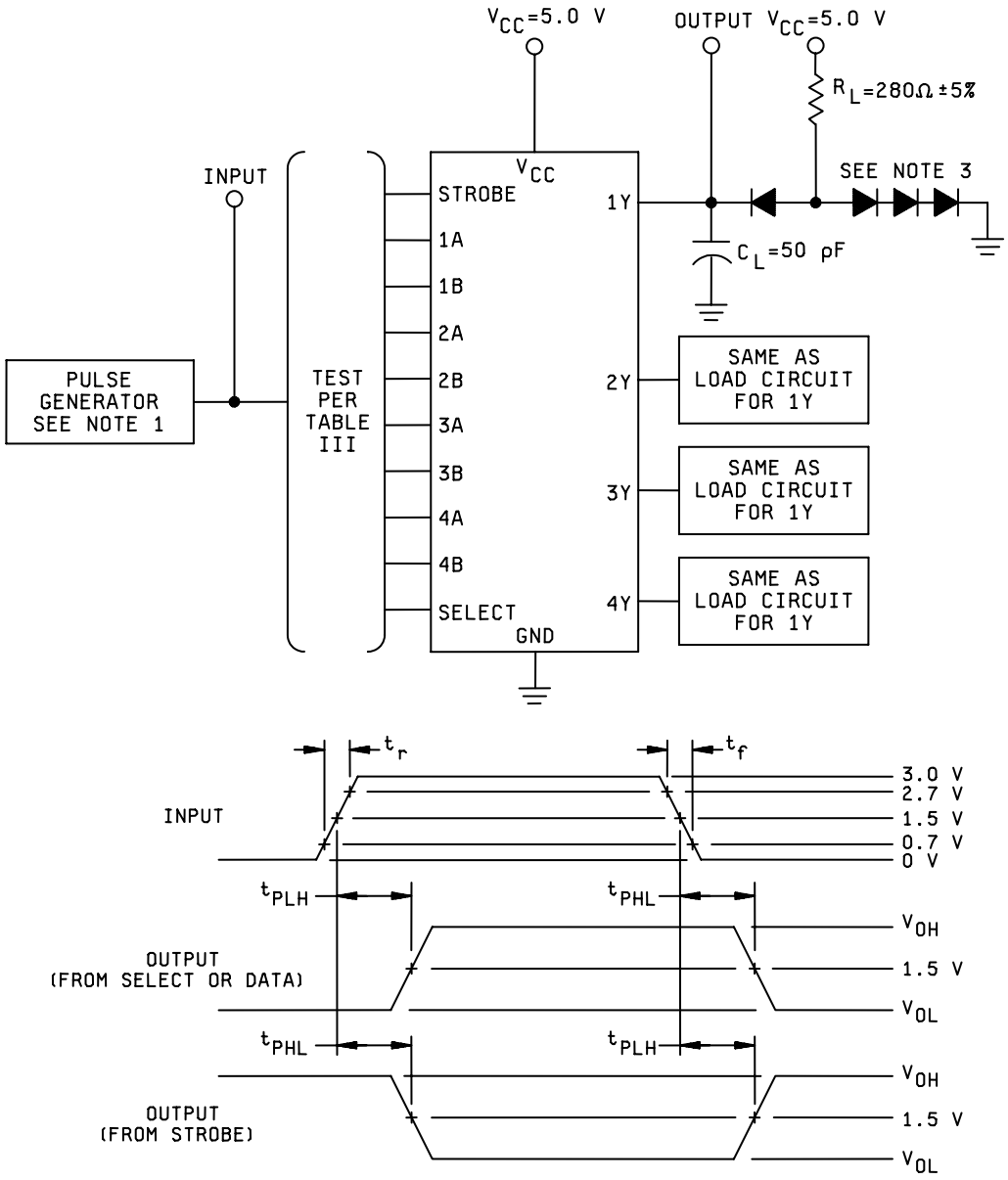



FIGURE 4. Switching time test circuits and waveforms for device type 02 – Continued.



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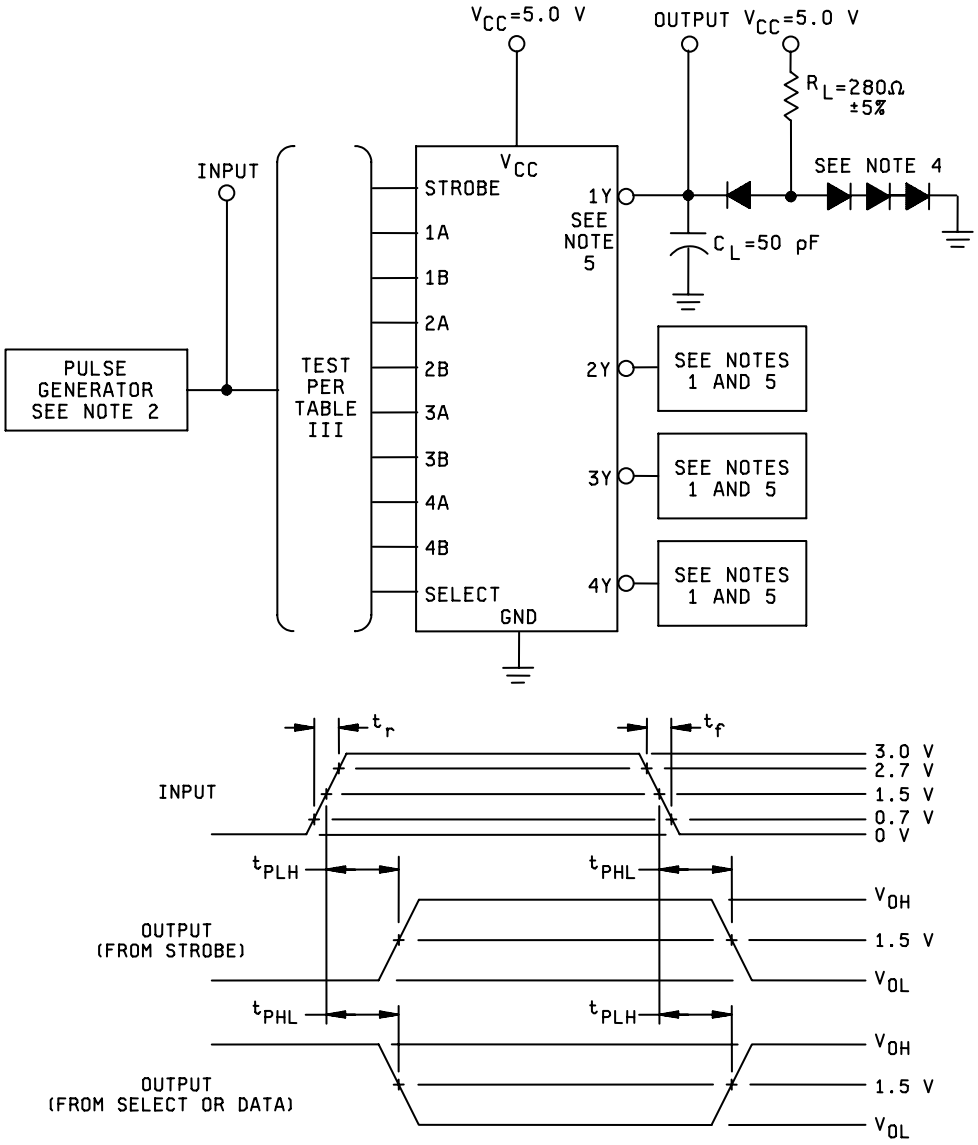



1. The input pulse has the following characteristics: PRR $\leq$ 1 MHz, $t_r = t_f \leq$ 2.5 ns, and $Z_{OUT} \approx$ 50 Ω.
2. $C_L$ includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Load circuit is required on a given output only where table III indicates “OUT” on that output. Load circuits may otherwise be omitted.

FIGURE 5. Switching time test circuits and waveforms for device type 03.



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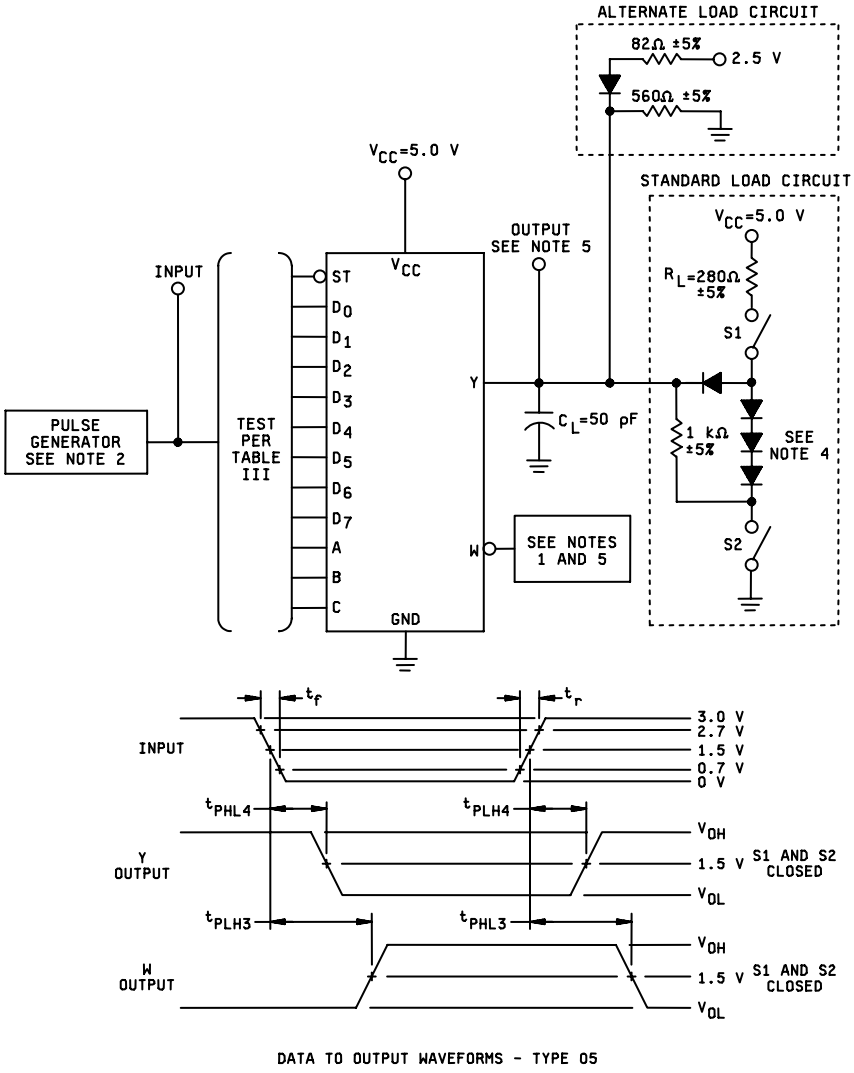

NOTES:

1. Connect same load as shown for output 1Y.
2. The input pulse has the following characteristics: PRR $\leq$ 1 MHz, $t_r = t_f \leq$ 2.5 ns, and $Z_{OUT} \approx$ 50 Ω.
3. $C_L$ includes probe and jig capacitance.
4. All diodes are 1N3064 or equivalent.
5. Load circuit is required on a given output only where table III indicates “OUT” on that output. Load circuits may otherwise be omitted.

FIGURE 6. Switching time test circuits and waveforms for device type 04.



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DATA TO OUTPUT WAVEFORMS - TYPE 05

NOTES:

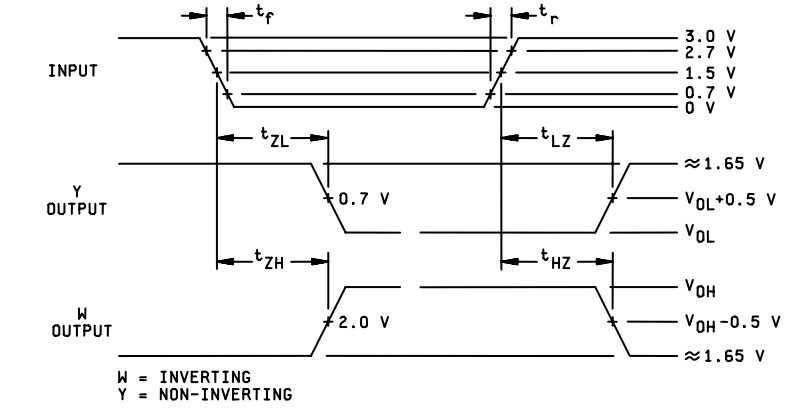
1. Connect same load as shown for Y output.
2. The input pulse has the following characteristics: $t_r = t_f \leq 2.5$ ns, PRR $\leq$ 1 MHz, and $Z_{OUT} \approx 50\ \Omega$.
3. $C_L$ includes probe and jig capacitance.
4. All diodes are 1N3064 or equivalent.
5. Load circuit is required on a given output only where table III indicates “OUT” on that output. Load circuits may otherwise be omitted.

   A. Output 1 is for an output with internal conditions such that the output is low except when disabled by the output control.
   B. Output 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
6. Manufacturer may test with either the standard load circuit or the alternate load circuit at his option.

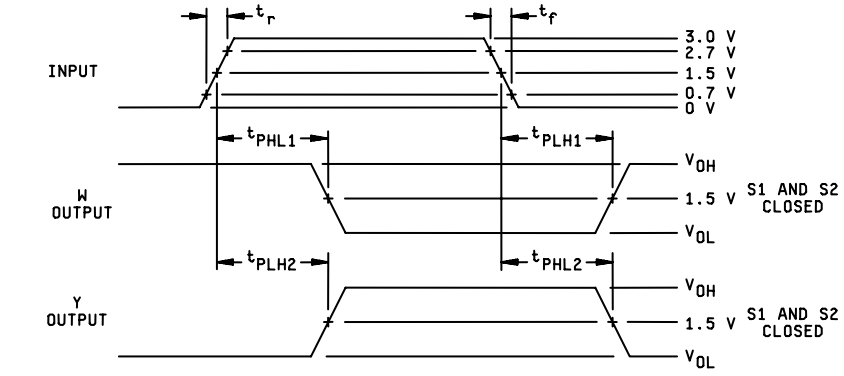
FIGURE 7. Switching time test circuits and waveforms for device type 05.



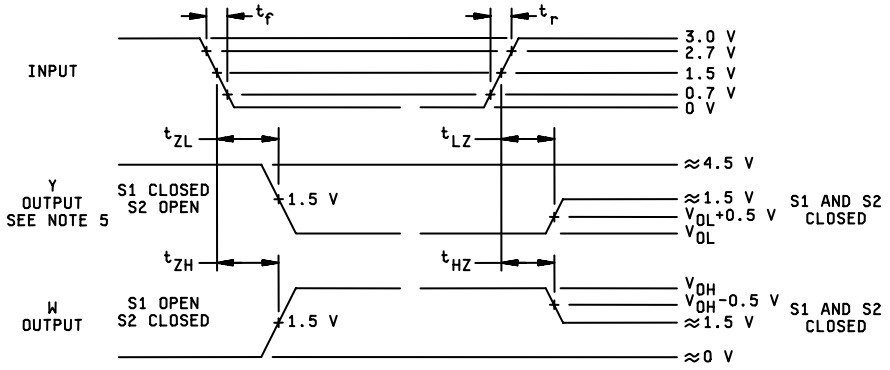
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:14Z
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WAVEFORMS FOR ALTERNATE LOADS - TYPE 05




SELECT TO OUTPUT WAVEFORMS - TYPE 05

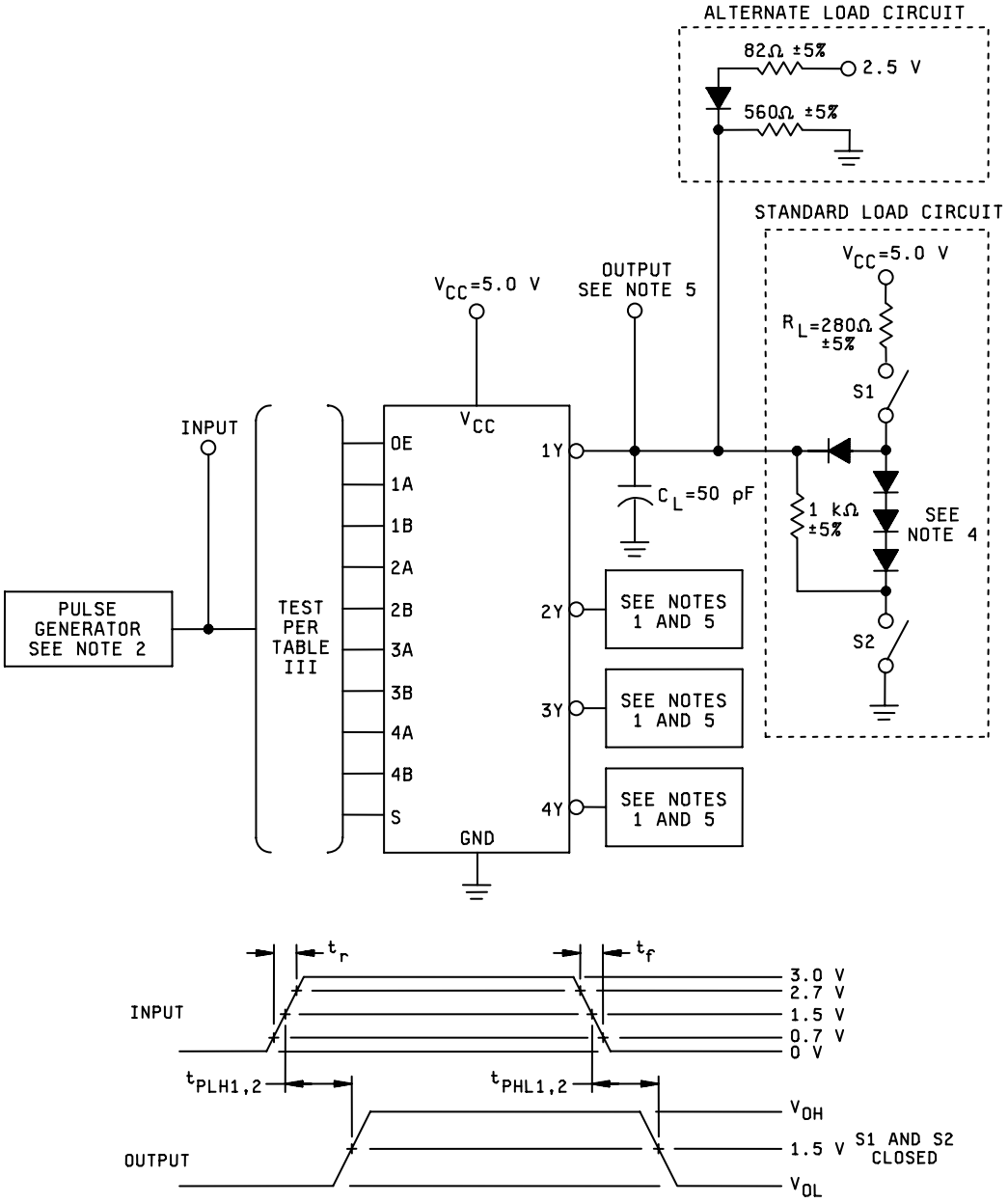



STROBE TO OUTPUT WAVEFORMS - TYPE 05

FIGURE 7. Switching time test circuits and waveforms for device type 05 – Continued.



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NOTES:

1. Connect same load as shown for 1Y output.
2. The input pulse has the following characteristics: $t_r = t_f \leq 2.5$ ns, PRR $\leq$ 1 MHz, and $Z_{OUT} \approx 50\ \Omega$.
3. $C_L$ includes probe and jig capacitance.
4. All diodes are 1N3064 or equivalent.
5. Load circuit is required on a given output only where table III indicates “OUT” on that output. Load circuits may otherwise be omitted.

   A. Output 1 is for an output with internal conditions such that the output is low except when disabled by the output control.
   B. Output 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
6. Manufacturer may test with either the standard load circuit or the alternate load circuit at his option.

FIGURE 8. Switching time test circuits and waveforms for device type 06.



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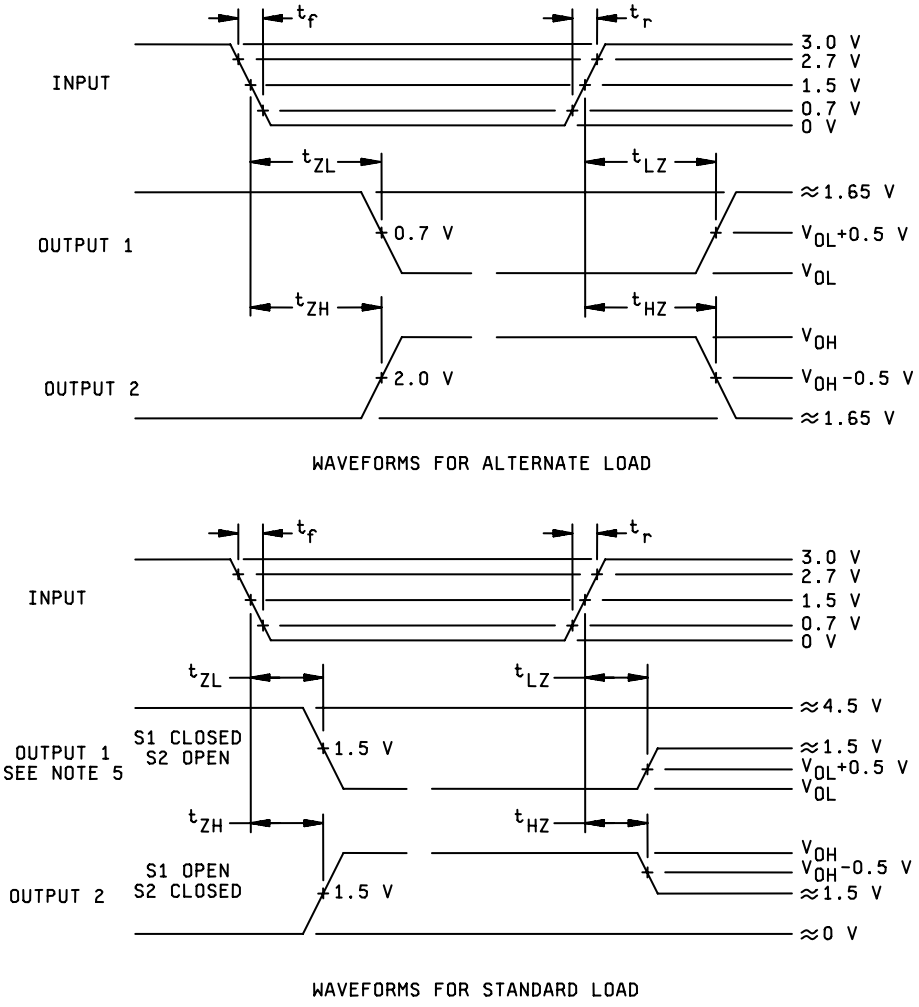
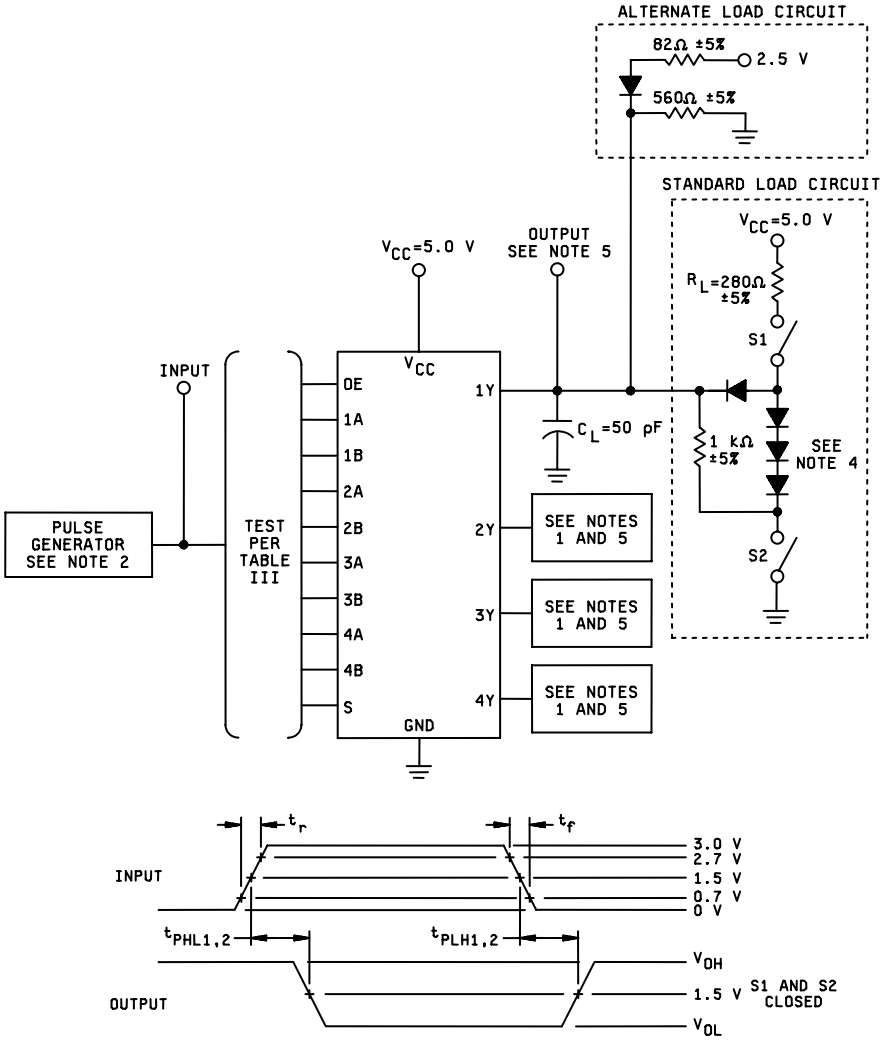



FIGURE 8. Switching time test circuits and waveforms for device type 06 - Continued.



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NOTES:

1. Connect same load as shown for 1Y output.
2. The input pulse has the following characteristics: $t_r = t_f \leq 2.5$ ns, PRR $\leq$ 1 MHz, and $Z_{OUT} \approx 50\ \Omega$.
3. $C_L$ includes probe and jig capacitance.
4. All diodes are 1N3064 or equivalent.
5. Load circuit is required on a given output only where table III indicates “OUT” on that output. Load circuits may otherwise be omitted.

   A. Output 1 is for an output with internal conditions such that the output is low except when disabled by the output control.
   B. Output 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
6. Manufacturer may test with either the standard load circuit or the alternate load circuit at his option.

FIGURE 9. Switching time test circuits and waveforms for device type 07.



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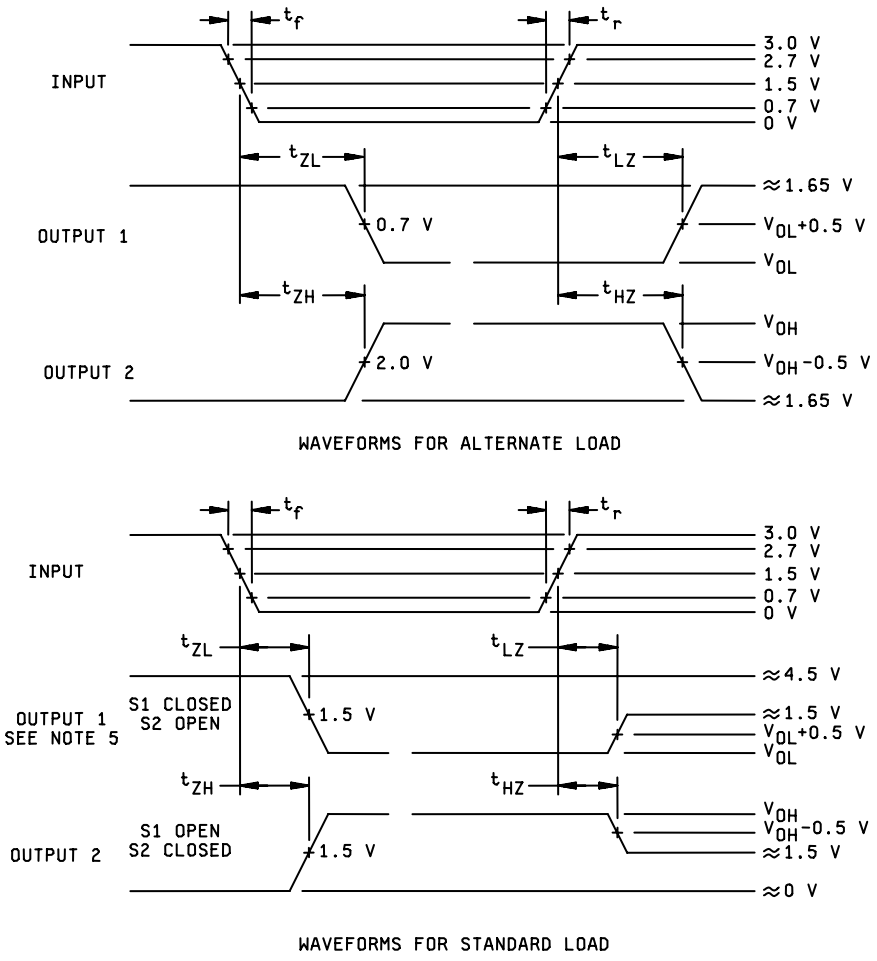
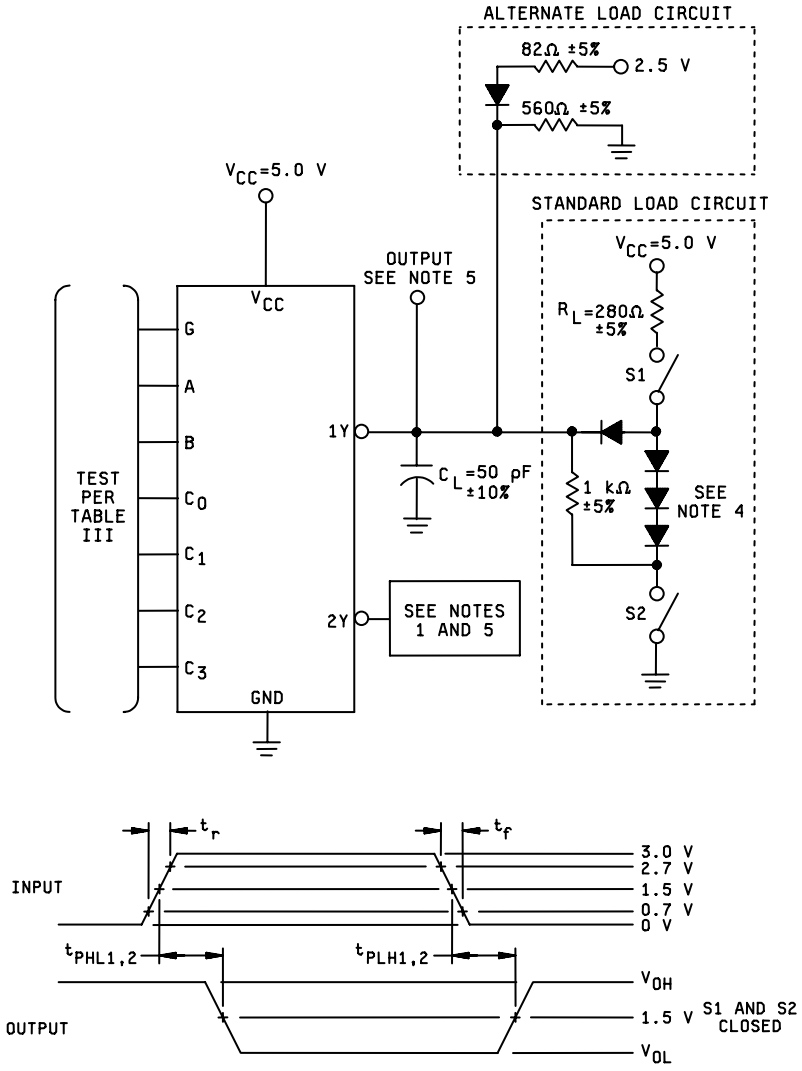



FIGURE 9. Switching time test circuits and waveforms for device type 07 - Continued.



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NOTES:

1. Connect same load as shown for 1Y output.
2. The input pulse has the following characteristics: $t_r = t_f \leq 2.5$ ns, PRR $\leq$ 1 MHz, and $Z_{OUT} \approx 50\ \Omega$.
3. $C_L$ includes probe and jig capacitance.
4. All diodes are 1N3064 or equivalent.
5. Load circuit is required on a given output only where table III indicates “OUT” on that output. Load circuits may otherwise be omitted.

   A. Output 1 is for an output with internal conditions such that the output is low except when disabled by the output control.
   B. Output 2 is for an output with internal conditions such that the output is high except when disabled by the output control.

6. Manufacturer may test with either the standard load circuit or the alternate load circuit at his option.

FIGURE 10. Switching time test circuits and waveforms for device type 08.



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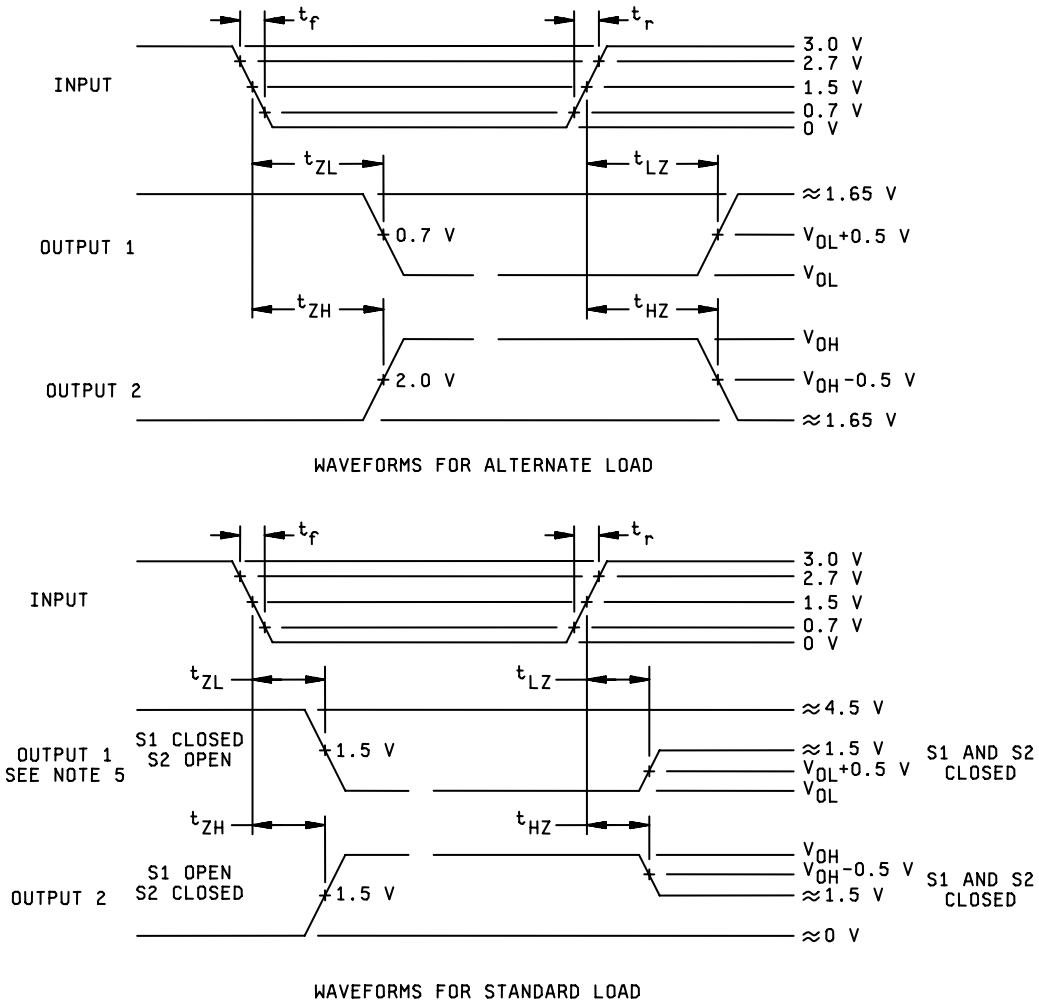



FIGURE 10. Switching time test circuits and waveforms for device type 08 - Continued.



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TABLE III. Group A inspection for device type 01

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | D3 | D2 | D1 | D0 | Y | W | ST | GND | C | B | A | D7 | D6 | D5 | D4 | $V_{CC}$ | | | | |
| 1<br>$T_C$ = +25°C | $V_{OL}$ | 3007 | 1 | | | | 2.0 V | | 20 mA | 0.8 V | GND | 0.8 V | 0.8 V | 0.8 V | | | | | 4.5 V | W | | 0.5 | V |
| | $V_{OL}$ | 3007 | 2 | | | | | 20 mA | | " | " | 2.0 V | 2.0 V | 2.0 V | 0.8 V | | | | " | Y | | 0.5 | " |
| | $V_{OH}$ | 3006 | 3 | | | | | | -1 mA | " | " | 2.0 V | 2.0 V | 2.0 V | 0.8 V | | | | " | W | 2.5 | | " |
| | $V_{OH}$ | 3006 | 4 | | | | 2.0 V | -1 mA | | " | " | 0.8 V | 0.8 V | 0.8 V | | | | | " | Y | 2.5 | | " |
| | $I_{OS}$ | 3011 | 5 | | | | | | GND | GND | " | 5.5 V | 5.5 V | 5.5 V | GND | | | | 5.5 V | W | -40 | -100 1/ | mA |
| | $I_{OS}$ | 3011 | 6 | | | | 5.5 V | GND | | GND | " | GND | GND | GND | | | | | " | Y | -40 | -100 1/ | " |
| | $I_{IH1}$ | 3010 | 7 | 2.7 V | | | | | | 5.5 V | " | 5.5 V | " | GND | | | | | " | D3 | | 50 | µA |
| | " | " | 8 | | 2.7 V | | | | | " | " | " | " | 5.5 V | | | | | " | D2 | | " | " |
| | " | " | 9 | | | 2.7 V | | | | " | " | " | 5.5 V | GND | | | | | " | D1 | | " | " |
| | " | " | 10 | | | | 2.7 V | | | " | " | " | 5.5 V | 5.5 V | | | | | " | D0 | | " | " |
| | " | " | 11 | | | | | | | 2.7 V | " | | | | | | | | " | ST | | " | " |
| | " | " | 12 | | | | | | | | " | 2.7 V | | | | | | | " | C | | " | " |
| | " | " | 13 | | | | | | | | " | | 2.7 V | | | | | | " | B | | " | " |
| | " | " | 14 | | | | | | | | " | | | 2.7 V | | | | | " | A | | " | " |
| | " | " | 15 | | | | | | | 5.5 V | " | GND | GND | GND | 2.7 V | | | | " | D7 | | " | " |
| | " | " | 16 | | | | | | | " | " | " | GND | 5.5 V | | 2.7 V | | | " | D6 | | " | " |
| | " | " | 17 | | | | | | | " | " | " | 5.5 V | GND | | | 2.7 V | | " | D5 | | " | " |
| | " | " | 18 | | | | | | | " | " | " | 5.5 V | 5.5 V | | | | 2.7 V | " | D4 | | " | " |
| | $I_{IH2}$ | " | 19 | 5.5 V | | | | | | " | " | 5.5 V | GND | GND | | | | | " | D3 | | 1.0 | mA |
| | " | " | 20 | | 5.5 V | | | | | " | " | " | GND | 5.5 V | | | | | " | D2 | | " | " |
| | " | " | 21 | | | 5.5 V | | | | " | " | " | 5.5 V | GND | | | | | " | D1 | | " | " |
| | " | " | 22 | | | | 5.5 V | | | " | " | " | 5.5 V | 5.5 V | | | | | " | D0 | | " | " |
| | " | " | 23 | | | | | | | " | " | | | | | | | | " | ST | | " | " |
| | " | " | 24 | | | | | | | | " | 5.5 V | | | | | | | " | C | | " | " |
| | " | " | 25 | | | | | | | | " | | 5.5 V | | | | | | " | B | | " | " |
| | " | " | 26 | | | | | | | | " | | | 5.5 V | | | | | " | A | | " | " |
| | " | " | 27 | | | | | | | 5.5 V | " | GND | GND | GND | 5.5 V | | | | " | D7 | | " | " |
| | " | " | 28 | | | | | | | " | " | " | GND | 5.5 V | | 5.5 V | | | " | D6 | | " | " |
| | " | " | 29 | | | | | | | " | " | " | 5.5 V | GND | | | 5.5 V | | " | D5 | | " | " |
| | " | " | 30 | | | | | | | " | " | " | " | 5.5 V | | | | 5.5 V | " | D4 | | " | " |
| | $I_{IL1}$ | 3009 | 31 | 0.5 V | | | | | | GND | " | " | " | 5.5 V | | | | | " | D3 | -1 | -2 | " |
| | " | " | 32 | | 0.5 V | | | | | " | " | " | " | GND | | | | | " | D2 | " | " | " |
| | " | " | 33 | | | 0.5 V | | | | " | " | " | GND | 5.5 V | | | | | " | D1 | " | " | " |
| | " | " | 34 | | | | 0.5 V | | | " | " | " | GND | GND | | | | | " | D0 | " | " | " |
| | " | " | 35 | | | | | | | 0.5 V | " | | | | | | | | " | ST | " | " | " |
| | " | " | 36 | | | | | | | | " | 0.5 V | | | | | | | " | C | " | " | " |
| | " | " | 37 | | | | | | | | " | | 0.5 V | | | | | | " | B | " | " | " |
| | " | " | 38 | | | | | | | | " | | | 0.5 V | | | | | " | A | " | " | " |
| | " | " | 39 | | | | | | | GND | " | 5.5 V | 5.5 V | 5.5 V | 0.5 V | | | | " | D7 | " | " | " |
| | " | " | 40 | | | | | | | " | " | " | 5.5 V | GND | | 0.5 V | | | " | D6 | " | " | " |
| | " | " | 41 | | | | | | | " | " | " | GND | 5.5 V | | | 0.5 V | | " | D5 | " | " | " |
| | " | " | 42 | | | | | | | " | " | " | " | GND | | | | 0.5 V | " | D4 | " | " | " |

See footnotes at end of device type 01.





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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | D3 | D2 | D1 | D0 | Y | W | ST | GND | C | B | A | D7 | D6 | D5 | D4 | $V_{CC}$ | | | | |
| 1 $T_C$ = +25°C (cont') | $I_{CC}$ | 3005 | 43 | GND | GND | GND | GND | | | GND | GND | GND | GND | GND | GND | GND | GND | GND | 5.5 V | $V_{CC}$ | | 70 | mA |
| | $I_{CEX}$ | | 44 | | | | | | 5.5 V | " | " | 5.5 V | 5.5 V | 5.5 V | GND | | | | 5.5 V | W | | 250 | µA |
| | | | 45 | | | | 5.5 V | 5.5 V | | " | " | GND | GND | GND | | | | | 5.5 V | Y | | 250 | µA |
| | $V_{IC}$ | | 46 | -18 mA | | | | | | | " | | | | | | | | 4.5 V | D3 | | -1.2 | V |
| | | | 47 | | -18 mA | | | | | | " | | | | | | | | " | D2 | | " | " |
| | | | 48 | | | -18 mA | | | | | " | | | | | | | | " | D1 | | " | " |
| | | | 49 | | | | -18 mA | | | | " | | | | | | | | " | D0 | | " | " |
| | | | 50 | | | | | | | -18 mA | " | | | | | | | | " | ST | | " | " |
| | | | 51 | | | | | | | | " | -18 mA | | | | | | | " | C | | " | " |
| | | | 52 | | | | | | | | " | | -18 mA | | | | | | " | B | | " | " |
| | | | 53 | | | | | | | | " | | | -18 mA | | | | | " | A | | " | " |
| | | | 54 | | | | | | | | " | | | | -18 mA | | | | " | D7 | | " | " |
| | | | 55 | | | | | | | | " | | | | | -18 mA | | | " | D6 | | " | " |
| | | | 56 | | | | | | | | " | | | | | | -18 mA | | " | D5 | | " | " |
| | | | 57 | | | | | | | | " | | | | | | | -18 mA | " | D4 | | " | " |
| 2 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. $V_{IL}$ = 0.7 V, $V_{OL}$ (max) = 0.45 V. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 7 | Truth | 3014 | 58 | A 2/ | A 2/ | A 2/ | A 2/ | L | H | A 2/ | GND | A 2/ | A 2/ | A 2/ | A 2/ | A 2/ | A 2/ | A 2/ | 4.5 V | 3/ | | | |
| $T_C$ = +25°C | table | " | 59 | A | A | A | B | L | H | B | " | B | B | B | A | A | A | A | " | " | | | |
| | test | " | 60 | B | B | B | A | H | L | " | " | " | " | B | B | B | B | B | " | " | | | |
| | | " | 61 | A | A | B | A | L | H | " | " | " | " | A | A | A | A | A | " | " | | | |
| | | " | 62 | B | B | A | B | H | L | " | " | " | " | A | B | B | B | B | " | " | | | |
| | | " | 63 | A | B | A | A | L | H | " | " | " | A | B | A | A | A | A | " | " | | | |
| | | " | 64 | B | A | B | B | H | L | " | " | " | " | B | B | B | B | B | " | " | | | |
| | | " | 65 | B | A | A | A | L | H | " | " | " | " | A | A | A | A | A | " | " | | | |
| | | " | 66 | A | B | B | B | H | L | " | " | " | " | A | B | B | B | B | " | " | | | |
| | | " | 67 | A | A | A | A | L | H | " | " | A | B | B | A | A | A | B | " | " | | | |
| | | " | 68 | B | B | B | B | H | L | " | " | " | " | B | B | B | B | A | " | " | | | |
| | | " | 69 | A | A | A | A | L | H | " | " | " | " | A | A | A | B | A | " | " | | | |
| | | " | 70 | B | B | B | B | H | L | " | " | " | " | A | B | B | A | B | " | " | | | |
| | | " | 71 | A | A | A | A | L | H | " | " | " | A | B | A | B | A | A | " | " | | | |
| | | " | 72 | B | B | B | B | H | L | " | " | " | " | B | B | A | B | B | " | " | | | |
| | | " | 73 | A | A | A | A | L | H | " | " | " | " | A | B | A | A | A | " | " | | | |
| | | " | 74 | B | B | B | B | H | L | " | " | " | " | A | A | B | B | B | " | " | | | |

See footnotes at end of device type 01.





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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | D3 | D2 | D1 | D0 | Y | W | ST | GND | C | B | A | D7 | D6 | D5 | D4 | $V_{CC}$ | | | | |
| 8 | Repeat subgroup 7 at $T_C$ = +125°C and -55°C. | | | | | | | | | | | | | | | | | | | | | | |
| 9 | $t_{PHL1}$ | 3003 | 75 | | | 2.7 V | GND | | OUT | GND | GND | GND | GND | IN | | | | | 5.0 V | W | 2.0 | 15.5 | ns |
| $T_C$ = +25°C | | Fig. 4 | 76 | | 2.7 V | | " | | " | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 77 | | | | " | | " | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PLH1}$ | " | 78 | | | 2.7 V | " | | " | " | " | GND | GND | IN | | | | | " | " | " | 17.0 | " |
| | | " | 79 | | 2.7 V | | " | | " | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 80 | | | | " | | " | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PHL2}$ | " | 81 | | | 2.7 V | " | OUT | | " | " | GND | GND | IN | | | | | " | Y | " | 20 | " |
| | | " | 82 | | 2.7 V | | " | " | | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 83 | | | | " | " | | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PLH2}$ | " | 84 | | | 2.7 V | " | " | | " | " | GND | GND | IN | | | | | " | " | " | " | " |
| | | " | 85 | | 2.7 V | | " | " | | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 86 | | | | " | " | | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PHL3}$ | " | 87 | | | | 2.7 V | | OUT | IN | " | GND | " | " | | | | | " | W | " | 14.0 | " |
| | $t_{PLH3}$ | " | 88 | | | | " | | OUT | " | " | " | " | " | | | | | " | W | " | 15.0 | " |
| | $t_{PHL4}$ | " | 89 | | | | " | OUT | | " | " | " | " | " | | | | | " | Y | " | 20 | " |
| | $t_{PLH4}$ | " | 90 | | | | " | OUT | | " | " | " | " | " | | | | | " | Y | " | 18.5 | " |
| | $t_{PHL5}$ | " | 91 | | | IN | IN | | OUT | GND | " | " | " | " | | | | | " | W | " | 9.0 | " |
| | | " | 92 | | | IN | | | " | " | " | " | " | 2.7 V | | | | | " | " | " | " | " |
| | | " | 93 | | IN | | | | " | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 94 | IN | | | | | " | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 95 | | | | | | " | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 96 | | | | | | " | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 97 | | | | | | " | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 98 | | | | | | " | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| | $t_{PLH5}$ | " | 99 | | | IN | IN | | " | " | " | GND | GND | GND | | | | | " | " | " | " | " |
| | | " | 100 | | | IN | | | " | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 101 | | IN | | | | " | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 102 | IN | | | | | " | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 103 | | | | | | " | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 104 | | | | | | " | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 105 | | | | | | " | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 106 | | | | | | " | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |

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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | D3 | D2 | D1 | D0 | Y | W | ST | GND | C | B | A | D7 | D6 | D5 | D4 | $V_{CC}$ | | | | |
| 9 | $t_{PHL6}$ | 3003 | 107 | | | | IN | OUT | | GND | GND | GND | GND | GND | | | | | 5.0 V | Y | 2.0 | 14.0 | ns |
| $T_C$ = +25°C | | Fig. 4 | 108 | | | IN | | " | | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 109 | | IN | | | " | | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 110 | IN | | | | " | | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 111 | | | | | " | | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 112 | | | | | " | | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 113 | | | | | " | | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 114 | | | | | " | | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| | $t_{PLH6}$ | " | 115 | | | | IN | " | | " | " | GND | GND | GND | | | | | " | " | " | " | " |
| | | " | 116 | | | IN | | " | | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 117 | | IN | | | " | | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 118 | IN | | | | " | | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 119 | | | | | " | | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 120 | | | | | " | | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 121 | | | | | " | | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 122 | | | | | " | | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| 10 | $t_{PHL1}$ | " | 123 | | | 2.7 V | GND | | OUT | " | " | GND | GND | IN | | | | | " | W | " | 20 | " |
| $T_C$ = +125°C | | " | 124 | | 2.7 V | | " | | " | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 125 | | | | " | | " | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PLH1}$ | " | 126 | | | 2.7 V | " | | " | " | " | GND | GND | IN | | | | | " | " | " | 22 | " |
| | | " | 127 | | 2.7 V | | " | | " | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 128 | | | | " | | " | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PHL2}$ | " | 129 | | | 2.7 V | " | OUT | | " | " | GND | GND | IN | | | | | " | Y | " | 26 | " |
| | | " | 130 | | 2.7 V | | " | " | | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 131 | | | | " | " | | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PLH2}$ | " | 132 | | | 2.7 V | " | " | | " | " | GND | GND | IN | | | | | " | " | " | " | " |
| | | " | 133 | | 2.7 V | | " | " | | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 134 | | | | " | " | | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PHL3}$ | " | 135 | | | | 2.7 V | | OUT | IN | " | GND | " | " | | | | | " | W | " | 18 | " |
| | $t_{PLH3}$ | " | 136 | | | | " | | OUT | " | " | " | " | " | | | | | " | W | " | 19.5 | " |
| | $t_{PHL4}$ | " | 137 | | | | " | OUT | | " | " | " | " | " | | | | | " | Y | " | 26 | " |
| | $t_{PLH4}$ | " | 138 | | | | " | OUT | | " | " | " | " | " | | | | | " | Y | " | 24 | " |

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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | D3 | D2 | D1 | D0 | Y | W | ST | GND | C | B | A | D7 | D6 | D5 | D4 | $V_{CC}$ | | | | |
| 10 | $t_{PHL5}$ | 3003 | 139 | | | | IN | | OUT | GND | GND | GND | GND | GND | | | | | 5.0 V | W | 2.0 | 11.5 | ns |
| $T_C$ = +125°C | | Fig. 4 | 140 | | | IN | | | " | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 141 | | IN | | | | " | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 142 | IN | | | | | " | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 143 | | | | | | " | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 144 | | | | | | " | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 145 | | | | | | " | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 146 | | | | | | " | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| | $t_{PLH5}$ | " | 147 | | | | IN | | " | " | " | GND | GND | GND | | | | | " | " | " | " | " |
| | | " | 148 | | | IN | | | " | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 149 | | IN | | | | " | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 150 | IN | | | | | " | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 151 | | | | | | " | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 152 | | | | | | " | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 153 | | | | | | " | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 154 | | | | | | " | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| | $t_{PHL6}$ | " | 155 | | | | IN | OUT | | " | " | GND | GND | GND | | | | | " | Y | " | 18 | " |
| | | " | 156 | | | IN | | " | | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 157 | | IN | | | " | | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 158 | IN | | | | " | | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 159 | | | | | " | | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 160 | | | | | " | | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 161 | | | | | " | | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 162 | | | | | " | | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| 10 | $t_{PLH6}$ | " | 163 | | | | IN | " | | " | " | GND | GND | GND | | | | | " | " | " | " | " |
| $T_C$ = +125°C | | " | 164 | | | IN | | " | | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 165 | | IN | | | " | | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 166 | IN | | | | " | | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 167 | | | | | " | | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 168 | | | | | " | | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 169 | | | | | " | | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 170 | | | | | " | | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| 11 | Same tests, terminal conditions, and limits as for subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

1/ For circuit B, $I_{OS(max)}$ = -110 mA.

2/ A = 2.4 V; B = 0.4 V.

3/ H ≥ 1.5 V; L ≤ 1.5 V.

4/ Only a summary of attributes is required.

5/ Case 2 pins not designated are NC.

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TABLE III. Group A inspection for device type 02
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | 1G | B | 1C3 | 1C2 | 1C1 | 1C0 | 1Y | GND | 2Y | 2C0 | 2C1 | 2C2 | 2C3 | A | 2G | $V_{CC}$ | | | | |
| 1 | $V_{OH}$ | 3006 | 1 | 0.8 V | 0.8 V | | | | 2.0 V | -1 mA | GND | | | | | | 0.8 V | | 4.5 V | 1Y | 2.5 | | V |
| $T_C$ = +25°C | $V_{OH}$ | 3006 | 2 | | 0.8 V | | | | | | " | -1 mA | 2.0 V | | | | 0.8 V | 0.8 V | " | 2Y | 2.5 | | " |
| | $V_{OL}$ | 3007 | 3 | 2.0 V | | | | | | 20 mA | " | | | | | | | | " | 1Y | | 0.5 | " |
| | $V_{OL}$ | 3007 | 4 | | | | | | | | " | 20 mA | | | | | | 2.0 V | " | 2Y | | 0.5 | " |
| | $V_{IC}$ | | 5 | | | | | | | | " | | | | | | -18 mA | | " | A | | -1.2 | " |
| | | | 6 | | -18 mA | | | | | | " | | | | | | | | " | B | | " | " |
| | | | 7 | | | | | | -18 mA | | " | | | | | | | | " | 1C0 | | " | " |
| | | | 8 | | | | | -18 mA | | | " | | | | | | | | " | 1C1 | | " | " |
| | | | 9 | | | | -18 mA | | | | " | | | | | | | | " | 1C2 | | " | " |
| | | | 10 | | | -18 mA | | | | | " | | | | | | | | " | 1C3 | | " | " |
| | | | 11 | -18 mA | | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | | 12 | | | | | | | | " | | -18 mA | | | | | | " | 2C0 | | " | " |
| | | | 13 | | | | | | | | " | | | -18 mA | | | | | " | 2C1 | | " | " |
| | | | 14 | | | | | | | | " | | | | -18 mA | | | | " | 2C2 | | " | " |
| | | | 15 | | | | | | | | " | | | | | -18 mA | | | " | 2C3 | | " | " |
| | | | 16 | | | | | | | | " | | | | | | | -18 mA | " | 2G | | " | " |
| | $I_{IL}$ | 3009 | 17 | | | | | | | | " | | | | | | 0.5 V | | 5.5 V | A | -1.0 | -2.0 | mA |
| | | " | 18 | | 0.5 V | | | | | | " | | | | | | | | " | B | " | " | " |
| | | " | 19 | 0.5 V | | | | | | | " | | | | | | | | " | 1G | " | " | " |
| | | " | 20 | | | | | | | | " | | | | | | | 0.5 V | " | 2G | " | " | " |
| | | " | 21 | GND | GND | | | | 0.5 V | | " | | | | | | GND | | " | 1C0 | " | " | " |
| | | " | 22 | " | GND | | | 0.5 V | | | " | | | | | | 5.5 V | | " | 1C1 | " | " | " |
| | | " | 23 | " | 5.5 V | | 0.5 V | | | | " | | | | | | GND | | " | 1C2 | " | " | " |
| | | " | 24 | " | 5.5 V | 0.5 V | | | | | " | | | | | | 5.5 V | | " | 1C3 | " | " | " |
| | | " | 25 | | GND | | | | | | " | | 0.5 V | | | | GND | GND | " | 2C0 | " | " | " |
| | | " | 26 | | GND | | | | | | " | | | 0.5 V | | | 5.5 V | " | " | 2C1 | " | " | " |
| | | " | 27 | | 5.5 V | | | | | | " | | | | 0.5 V | | GND | " | " | 2C2 | " | " | " |
| | | " | 28 | | 5.5 V | | | | | | " | | | | | 0.5 V | 5.5 V | " | " | 2C3 | " | " | " |
| | $I_{IH1}$ | 3010 | 29 | | | | | | | | " | | | | | | 2.7 V | | " | A | | 50 | µA |
| | | " | 30 | | 2.7 V | | | | | | " | | | | | | | | " | B | | " | " |
| | | " | 31 | 2.7 V | | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | " | 32 | | | | | | | | " | | | | | | | 2.7 V | " | 2G | | " | " |
| | | " | 33 | 5.5 V | 5.5 V | | | | 2.7 V | | " | | | | | | 5.5 V | | " | 1C0 | | " | " |
| | | " | 34 | " | 5.5 V | | | 2.7 V | | | " | | | | | | GND | | " | 1C1 | | " | " |
| | | " | 35 | " | GND | | 2.7 V | | | | " | | | | | | 5.5 V | | " | 1C2 | | " | " |
| | | " | 36 | " | GND | 2.7 V | | | | | " | | | | | | GND | | " | 1C3 | | " | " |
| | | " | 37 | | 5.5 V | | | | | | " | | 2.7 V | | | | 5.5 V | 5.5 V | " | 2C0 | | " | " |
| | | " | 38 | | 5.5 V | | | | | | " | | | 2.7 V | | | GND | " | " | 2C1 | | " | " |
| | | " | 39 | | GND | | | | | | " | | | | 2.7 V | | 5.5 V | " | " | 2C2 | | " | " |
| | | " | 40 | | GND | | | | | | " | | | | | 2.7 V | GND | " | " | 2C3 | " | " | " |

See footnotes at end of device type 02



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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | 1G | B | 1C3 | 1C2 | 1C1 | 1C0 | 1Y | GND | 2Y | 2C0 | 2C1 | 2C2 | 2C3 | A | 2G | $V_{CC}$ | | | | |
| 1 | $I_{IH2}$ | 3010 | 41 | | | | | | | | GND | | | | | | 5.5 V | | 5.5 V | A | | 1.0 | mA |
| $T_C$ = +25°C | | " | 42 | | 5.5 V | | | | | | " | | | | | | | | " | B | | " | " |
| | | " | 43 | 5.5 V | | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | " | 44 | | | | | | | | " | | | | | | | 5.5 V | " | 2G | | " | " |
| | | " | 45 | 5.5 V | 5.5 V | | | | 5.5 V | | " | | | | | | 5.5 V | | " | 1C0 | | " | " |
| | | " | 46 | " | 5.5 V | | | 5.5 V | | | " | | | | | | GND | | " | 1C1 | | " | " |
| | | " | 47 | " | GND | | 5.5 V | | | | " | | | | | | 5.5 V | | " | 1C2 | | " | " |
| | | " | 48 | " | GND | 5.5 V | | | | | " | | | | | | GND | | " | 1C3 | | " | " |
| | | " | 49 | | 5.5 V | | | | | | " | | 5.5 V | | | | 5.5 V | 5.5 V | " | 2C0 | | " | " |
| | $I_{IH2}$ | " | 50 | | 5.5 V | | | | | | " | | | 5.5 V | | | GND | " | " | 2C1 | | 1.0 | mA |
| | | " | 51 | | GND | | | | | | " | | | | 5.5 V | | 5.5 V | " | " | 2C2 | | " | " |
| | | " | 52 | | " | | | | | | " | | | | | 5.5 V | GND | " | " | 2C3 | | " | " |
| | $I_{OS}$ | 3011 | 53 | GND | " | | | | 5.5 V | GND | " | | | | | | " | | " | 1Y | -40 | -100 1/ | " |
| | | " | 54 | " | " | | | | | | " | GND | 5.5 V | | | | " | GND | " | 2Y | -40 | -100 1/ | " |
| | $I_{CC}$ | 3005 | 55 | " | " | GND | GND | GND | GND | | " | | GND | GND | GND | GND | " | GND | " | $V_{CC}$ | | 70 | " |
| | $I_{CEX}$ | | 56 | GND | " | | | | 5.5 V | 5.5 V | " | | | | | | " | | " | 1Y | | 250 | µA |
| | | | 57 | | " | | | | | | " | 5.5 V | 5.5 V | | | | " | GND | " | 2Y | | 250 | µA |
| 2 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. $V_{IL}$ = 0.7 V, $V_{OL}$ = 0.45 V. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 7 2/ | Truth | 3014 | 58 | A 3/ | B 3/ | A 3/ | A 3/ | A 3/ | A 3/ | L | GND | L | A 2/ | A 2/ | A 2/ | A 2/ | B 2/ | A 2/ | 4.5 V | See 4/ | | | |
| $T_C$ = +25°C | table | " | 59 | B | " | A | A | A | B | L | " | L | B | A | A | A | " | B | " | " | | | |
| | test | " | 60 | " | " | B | B | B | A | H | " | H | A | B | B | B | " | " | " | " | | | |
| | | " | 61 | " | " | A | A | B | A | L | " | L | A | B | A | A | A | " | " | " | | | |
| | | " | 62 | " | " | B | B | A | B | H | " | H | B | A | B | B | A | " | " | " | | | |
| | | " | 63 | " | A | A | B | A | A | L | " | L | A | A | B | A | B | " | " | " | | | |
| | | " | 64 | " | " | B | A | B | B | H | " | H | B | B | A | B | B | " | " | " | | | |
| | | " | 65 | " | " | B | A | A | A | L | " | L | A | A | A | B | A | " | " | " | | | |
| | | " | 66 | " | " | A | B | B | B | H | " | H | B | B | B | A | A | " | " | " | | | |
| 8 | Same tests, terminal conditions, and limits as for subgroup 7, except $T_C$ = +125°C and -55°C. | | | | | | | | | | | | | | | | | | | | | | |

See footnotes at end of device type 02



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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | 1G | B | 1C3 | 1C2 | 1C1 | 1C0 | 1Y | GND | 2Y | 2C0 | 2C1 | 2C2 | 2C3 | A | 2G | $V_{CC}$ | | | | |
| 9 | $t_{PHL1}$ | 3003 | 67 | GND | GND | | | | IN | OUT | GND | | | | | | GND | | 5.0 V | 1Y | 2.0 | 11.0 | ns |
| $T_C$ = +25°C | | Fig. 5 | 68 | " | GND | | | IN | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 69 | " | 2.7 V | | IN | | | " | " | | | | | | GND | | " | " | " | " | " |
| | | " | 70 | " | 2.7 V | IN | | | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 71 | | GND | | | | | | " | OUT | IN | | | | GND | GND | " | 2Y | " | " | " |
| | | " | 72 | | GND | | | | | | " | " | | IN | | | 2.7 V | " | " | " | " | " | " |
| | | " | 73 | | 2.7 V | | | | | | " | " | | | IN | | GND | " | " | " | " | " | " |
| | | " | 74 | | 2.7 V | | | | | | " | " | | | | IN | 2.7 V | " | " | " | " | " | " |
| | $t_{PLH1}$ | " | 75 | GND | GND | | | | IN | OUT | " | | | | | | GND | | " | 1Y | " | " | " |
| | | " | 76 | " | GND | | | IN | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 77 | " | 2.7 V | | IN | | | " | " | | | | | | GND | | " | " | " | " | " |
| | | " | 78 | " | 2.7 V | IN | | | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 79 | | GND | | | | | | " | OUT | IN | | | | GND | GND | " | 2Y | " | " | " |
| | | " | 80 | | GND | | | | | | " | " | | IN | | | 2.7 V | " | " | " | " | " | " |
| | | " | 81 | | 2.7 V | | | | | | " | " | | | IN | | GND | " | " | " | " | " | " |
| | | " | 82 | | 2.7 V | | | | | | " | " | | | | IN | 2.7 V | " | " | " | " | " | " |
| | $t_{PHL2}$ | " | 83 | GND | GND | | | GND | 2.7 V | OUT | " | | | | | | IN | | " | 1Y | " | 20 | " |
| | | " | 84 | GND | IN | | GND | | 2.7 V | OUT | " | | | | | | GND | | " | 1Y | " | " | " |
| | | " | 85 | | GND | | | | | | " | OUT | 2.7 V | GND | | | IN | GND | " | 2Y | " | " | " |
| | | " | 86 | | IN | | | | | | " | OUT | 2.7 V | | GND | | GND | GND | " | 2Y | " | " | " |
| | $t_{PLH2}$ | " | 87 | GND | GND | | | GND | 2.7 V | OUT | GND | | | | | | IN | | " | 1Y | " | " | " |
| | | " | 88 | GND | IN | | GND | | 2.7 V | OUT | " | | | | | | GND | | " | 1Y | " | " | " |
| | | " | 89 | | GND | | | | | | " | OUT | 2.7 V | GND | | | IN | GND | " | 2Y | " | " | " |
| | | " | 90 | | IN | | | | | | " | OUT | 2.7 V | | GND | | GND | GND | " | 2Y | " | " | " |
| | $t_{PHL3}$ | " | 91 | IN | GND | | | | 2.7 V | OUT | " | | | | | | " | | " | 1Y | " | 15.5 | " |
| | | " | 92 | | " | | | | | | " | OUT | 2.7 V | | | | " | IN | " | 2Y | " | 15.5 | " |
| | $t_{PLH3}$ | " | 93 | IN | " | | | | 2.7 V | OUT | " | | | | | | " | | " | 1Y | " | 17.0 | " |
| | | " | 94 | | " | | | | | | " | OUT | 2.7 V | | | | " | IN | " | 2Y | " | 17.0 | " |

See footnotes at end of device type 02





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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | 1G | B | 1C3 | 1C2 | 1C1 | 1C0 | 1Y | GND | 2Y | 2C0 | 2C1 | 2C2 | 2C3 | A | 2G | $V_{CC}$ | | | | |
| 10 | $t_{PHL1}$ | 3003 | 95 | GND | GND | | | | IN | OUT | GND | | | | | | GND | | 5.0 V | 1Y | 2.0 | 14.5 | ns |
| $T_C$ = +125°C | | Fig. 5 | 96 | " | GND | | | IN | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 97 | " | 2.7 V | | IN | | | " | " | | | | | | GND | | " | " | " | " | " |
| | | " | 98 | " | 2.7 V | IN | | | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 99 | | GND | | | | | | " | OUT | IN | | | | GND | GND | " | 2Y | " | " | " |
| | | " | 100 | | GND | | | | | | " | " | | IN | | | 2.7 V | " | " | " | " | " | " |
| | | " | 101 | | 2.7 V | | | | | | " | " | | | IN | | GND | " | " | " | " | " | " |
| | | " | 102 | | 2.7 V | | | | | | " | " | | | | IN | 2.7 V | " | " | " | " | " | " |
| | $t_{PLH1}$ | " | 103 | GND | GND | | | | IN | OUT | " | | | | | | GND | | " | 1Y | " | 14.5 | " |
| | | " | 104 | " | GND | | | IN | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 105 | " | 2.7 V | | IN | | | " | " | | | | | | GND | | " | " | " | " | " |
| | | " | 106 | " | 2.7 V | IN | | | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 107 | | GND | | | | | | " | OUT | IN | | | | GND | GND | " | 2Y | " | " | " |
| | | " | 108 | | GND | | | | | | " | " | | IN | | | 2.7 V | " | " | " | " | " | " |
| | | " | 109 | | 2.7 V | | | | | | " | " | | | IN | | GND | " | " | " | " | " | " |
| | | " | 110 | | 2.7 V | | | | | | " | " | | | | IN | 2.7 V | " | " | " | " | " | " |
| | $t_{PHL2}$ | " | 111 | GND | GND | | | GND | 2.7 V | OUT | " | | | | | | IN | | " | 1Y | " | 26 | " |
| | | " | 112 | GND | IN | | GND | | 2.7 V | OUT | " | | | | | | GND | | " | 1Y | " | " | " |
| | | " | 113 | | GND | | | | | | " | OUT | 2.7 V | GND | | | IN | GND | " | 2Y | " | " | " |
| | | " | 114 | | IN | | | | | | " | OUT | 2.7 V | | GND | | GND | GND | " | 2Y | " | " | " |
| | $t_{PLH2}$ | " | 115 | GND | GND | | | GND | 2.7 V | OUT | " | | | | | | IN | | " | 1Y | " | 26 | " |
| | | " | 116 | GND | IN | | GND | | 2.7 V | OUT | " | | | | | | GND | | " | 1Y | " | " | " |
| | | " | 117 | | GND | | | | | | " | OUT | 2.7 V | GND | | | IN | GND | " | 2Y | " | " | " |
| | | " | 118 | | IN | | | | | | " | OUT | 2.7 V | | GND | | GND | GND | " | 2Y | " | " | " |
| | $t_{PHL3}$ | " | 119 | IN | GND | | | | 2.7 V | OUT | " | | | | | | " | IN | " | 1Y | " | 21 | " |
| | | " | 120 | | " | | | | | | " | OUT | 2.7 V | | | | " | | " | 2Y | " | 21 | " |
| | $t_{PLH3}$ | " | 121 | IN | " | | | | 2.7 V | OUT | " | | | | | | " | IN | " | 1Y | " | 22 | " |
| | | " | 122 | | " | | | | | | " | OUT | 2.7 V | | | | " | | " | 2Y | " | 22 | " |
| 11 | Same tests, terminal conditions, and limits as for subgroup 10 except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

1/ For circuit B, $I_{OS(max)}$ = -110 mA

2/ Only a summary of attributes is required.

3/ A = 2.4 V; B = 0.4 V.

4/ H ≥ 1.5 V; L ≤ 1.5 V.

5/ Case 2 pins not designated are NC.





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TABLE III. Group A inspection for device type 03
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | G | $V_{CC}$ | | | | |
| 1<br>$T_C$ = +25°C | $V_{OH}$ | 3006 | 1 | 2.0 V | | 2.0 V | -1 mA | | 2.0 V | -1 mA | GND | | | | | | | 0.8 V | 4.5 V | 1Y | 2.5 | | V |
| | | " | 2 | " | | | | | | | " | | | | | | | " | " | 2Y | " | | " |
| | | " | 3 | " | | | | | | | " | -1 mA | 2.0 V | | | | | " | " | 3Y | " | | " |
| | | " | 4 | " | | | | | | | " | | | | -1 mA | 2.0 V | | " | " | 4Y | " | | " |
| | $V_{OL}$ | 3007 | 5 | | | | 20 mA | | | | " | | | | | | | 2.0 V | " | 1Y | | 0.5 | " |
| | | " | 6 | | | | | | | 20 mA | " | | | | | | | " | " | 2Y | | " | " |
| | | " | 7 | | | | | | | | " | 20 mA | | | | | | " | " | 3Y | | " | " |
| | | " | 8 | | | | | | | | " | | | | 20 mA | | | " | " | 4Y | | " | " |
| | $V_{IC}$ | | 9 | -18 mA | | | | | | | " | | | | | | | | " | S | | -1.2 | " |
| | | | 10 | | -18 mA | | | | | | " | | | | | | | | " | 1A | | " | " |
| | | | 11 | | | -18 mA | | | | | " | | | | | | | | " | 1B | | " | " |
| | | | 12 | | | | | -18 mA | | | " | | | | | | | | " | 2A | | " | " |
| | | | 13 | | | | | | -18 mA | | " | | | | | | | | " | 2B | | " | " |
| | | | 14 | | | | | | | | " | | -18 mA | | | | | | " | 3B | | " | " |
| | | | 15 | | | | | | | | " | | | -18 mA | | | | | " | 3A | | " | " |
| | | | 16 | | | | | | | | " | | | | | -18 mA | | | " | 4B | | " | " |
| | | | 17 | | | | | | | | " | | | | | | -18 mA | | " | 4A | | " | " |
| | | | 18 | | | | | | | | " | | | | | | | -18 mA | " | G | | " | " |
| | $I_{IL}$ | 3009 | 19 | 5.5 V | | | | | | | " | | | | | | | 0.5 V | 5.5 V | G | -2 6/ | -4 6/ | mA |
| | | " | 20 | 0.5 V | | | | | | | " | | | | | | | 5.5 V | " | S | -2 6/ | -4 6/ | " |
| | | " | 21 | GND | 0.5 V | | | | | | " | | | | | | | GND | " | 1A | -1 7/ | -2 7/ | " |
| | | " | 22 | 5.5 V | | 0.5 V | | | | | " | | | | | | | " | " | 1B | " " | " " | " |
| | | " | 23 | GND | | | | 0.5 V | | | " | | | | | | | " | " | 2A | " " | " " | " |
| | | " | 24 | 5.5 V | | | | | 0.5 V | | " | | | | | | | " | " | 2B | " " | " " | " |
| | | " | 25 | 5.5 V | | | | | | | " | | 0.5 V | | | | | " | " | 3B | " " | " " | " |
| | | " | 26 | GND | | | | | | | " | | | 0.5 V | | | | " | " | 3A | " " | " " | " |
| | | " | 27 | 5.5 V | | | | | | | " | | | | | 0.5 V | | " | " | 4B | " " | " " | " |
| | | " | 28 | GND | | | | | | | " | | | | | | 0.5 V | " | " | 4A | " " | " " | " |
| | $I_{IH1}$ | 3010 | 29 | GND | | | | | | | " | | | | | | | 2.7 V | " | G | | 100 | µA |
| | | " | 30 | 2.7 V | | | | | | | " | | | | | | | GND | " | S | | 100 | " |
| | | " | 31 | | 2.7 V | | | | | | " | | | | | | | 5.5 V | " | 1A | | 50 | " |
| | | " | 32 | | | 2.7 V | | | | | " | | | | | | | " | " | 1B | | " | " |
| | | " | 33 | | | | | 2.7 V | | | " | | | | | | | " | " | 2A | | " | " |
| | | " | 34 | | | | | | 2.7 V | | " | | | | | | | " | " | 2B | | " | " |
| | | " | 35 | | | | | | | | " | | 2.7 V | | | | | " | " | 3B | | " | " |
| | | " | 36 | | | | | | | | " | | | 2.7 V | | | | " | " | 3A | | " | " |
| | | " | 37 | | | | | | | | " | | | | | 2.7 V | | " | " | 4B | | " | " |
| | | " | 38 | | | | | | | | " | | | | | | 2.7 V | " | " | 4A | | " | " |

See footnotes at end of device type 03



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TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | G | $V_{CC}$ | | | | |
| 1 | $I_{IH2}$ | 3010 | 39 | GND | | | | | | | GND | | | | | | | 5.5 V | 5.5 V | G | | 1.0 | mA |
| $T_C$ = +25°C | | " | 40 | 5.5 V | | | | | | | " | | | | | | | GND | " | S | | " | " |
| | | " | 41 | | 5.5 V | | | | | | " | | | | | | | 5.5 V | " | 1A | | " | " |
| | | " | 42 | | | 5.5 V | | | | | " | | | | | | | " | " | 1B | | " | " |
| | | " | 43 | | | | | 5.5 V | | | " | | | | | | | " | " | 2A | | " | " |
| | | " | 44 | | | | | | 5.5 V | | " | | | | | | | " | " | 2B | | " | " |
| | | " | 45 | | | | | | | | " | | 5.5 V | | | | | " | " | 3B | | " | " |
| | | " | 46 | | | | | | | | " | | | 5.5 V | | | | " | " | 3A | | " | " |
| | | " | 47 | | | | | | | | " | | | | 5.5 V | | | " | " | 4B | | " | " |
| | | " | 48 | | | | | | | | " | | | | | 5.5 V | | " | " | 4A | | " | " |
| | $I_{OS}$ | 3011 | 49 | 5.5 V | | 5.5 V | GND | | | | " | | | | | | | GND | " | 1Y | -40 | -100 1/ | " |
| | | 3011 | 50 | " | | | | | 5.5 V | GND | " | | | | | | | GND | " | 2Y | -40 | -100 1/ | " |
| | $I_{OS}$ | 3011 | 51 | " | | | | | | | " | GND | 5.5 V | | | | | GND | 5.5 V | 3Y | -40 | -100 1/ | mA |
| | | 3011 | 52 | " | | | | | | | " | | | | GND | 5.5 V | | GND | " | 4Y | -40 | -100 1/ | " |
| | $I_{CC}$ | 3005 | 53 | " | GND | GND | | GND | GND | | " | | GND | GND | | GND | GND | 5.5 V | " | $V_{CC}$ | | 78 | |
| | $I_{CEX}$ | | 54 | " | | 5.5 V | 5.5 V | | | | " | | | | | | | GND | " | 1Y | | 250 | µA |
| | | | 55 | " | | | | | 5.5 V | 5.5 V | " | | | | | | | " | " | 2Y | | " | " |
| | | | 56 | " | | | | | | | " | 5.5 V | 5.5 V | | | | | " | " | 3Y | | " | " |
| | | | 57 | " | | | | | | | " | | | | 5.5 V | 5.5 V | | " | " | 4Y | | " | " |
| 2 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. $V_{IL}$ = 0.7 V, $V_{OL(max)}$ = 0.45 V. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 7 4/ | Truth | 3014 | 58 | A 2/ | A 2/ | A 2/ | L | A 2/ | A 2/ | L | GND | L | A 2/ | A 2/ | L | A 2/ | A 2/ | A 2/ | 4.5 V | See 3/ | | | |
| $T_C$ = +25°C | table | " | 59 | " | A | B | L | | | | " | | | | | | | B | " | " | | | |
| | test | " | 60 | " | B | A | H | | | | " | | | | | | | " | " | " | | | |
| | | " | 61 | B | B | A | L | | | | " | | | | | | | " | " | " | | | |
| | | " | 62 | B | A | B | H | | | | " | | | | | | | " | " | " | | | |
| | | " | 63 | A | | | | A | B | L | " | | | | | | | " | " | " | | | |
| | | " | 64 | A | | | | B | A | H | " | | | | | | | " | " | " | | | |
| | | " | 65 | B | | | | B | A | L | " | | | | | | | " | " | " | | | |
| | | " | 66 | B | | | | A | B | H | " | | | | | | | " | " | " | | | |
| | | " | 67 | A | | | | | | | " | L | B | A | | | | " | " | " | | | |
| | | " | 68 | A | | | | | | | " | H | A | B | | | | " | " | " | | | |
| | | " | 69 | B | | | | | | | " | L | A | B | | | | " | " | " | | | |
| | | " | 70 | B | | | | | | | " | H | B | A | | | | " | " | " | | | |
| | | " | 71 | A | | | | | | | " | | | | L | B | A | " | " | " | | | |
| | | " | 72 | A | | | | | | | " | | | | H | A | B | " | " | " | | | |
| | | " | 73 | B | | | | | | | " | | | | L | A | B | " | " | " | | | |
| | | " | 74 | B | | | | | | | " | | | | H | B | A | " | " | " | | | |

See footnotes at end of device type 03



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TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | G | $V_{CC}$ | | | | |
| 8 4/ | Same tests, terminal conditions, and limits as for subgroup 7, except $T_C$ = +125°C and -55°C. | | | | | | | | | | | | | | | | | | | | | | |
| 9 | $t_{PHL1}$ | 3003 | 75 | IN | GND | 2.7 V | OUT | | | | GND | | | | | | | GND | 5.0 V | 1Y | 2.0 | 14 | ns |
| $T_C$ = +25°C | | Fig. 6 | 76 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 77 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 78 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PLH1}$ | " | 79 | " | GND | 2.7 V | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 80 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 81 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 82 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PHL2}$ | " | 83 | GND | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 9.0 | " |
| | | " | 84 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 85 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 86 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PLH2}$ | " | 87 | " | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 88 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 89 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 90 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PHL3}$ | " | 91 | " | 2.7 V | GND | OUT | | | | " | | | | | | | IN | 5.0 V | 1Y | " | 14 | " |
| | | " | 92 | " | | | | 2.7 V | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 93 | " | | | | | | | " | OUT | GND | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 94 | " | | | | | | | " | | | | OUT | GND | 2.7 V | " | " | 4Y | " | " | " |
| | $t_{PLH3}$ | " | 95 | " | 2.7 V | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 13.5 | " |
| | | " | 96 | " | | | | 2.7 V | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 97 | " | | | | | | | " | OUT | GND | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 98 | " | | | | | | | " | | | | OUT | GND | 2.7 V | " | " | 4Y | " | " | " |

See footnotes at end of device type 03.



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TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | G | $V_{CC}$ | | | | |
| 10 | $t_{PHL1}$ | 3003 | 99 | IN | GND | 2.7 V | OUT | | | | GND | | | | | | | GND | 5.0 V | 1Y | 2.0 | 18.5 | ns |
| $T_C$ = +125°C | | Fig. 6 | 100 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 101 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 102 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PLH1}$ | " | 103 | " | GND | 2.7 V | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 104 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 105 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 106 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PHL2}$ | " | 107 | GND | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 12 | " |
| | | " | 108 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 109 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 110 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PLH2}$ | " | 111 | " | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 112 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 113 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 114 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PHL3}$ | " | 115 | " | 2.7 V | GND | OUT | | | | " | | | | | | | IN | " | 1Y | " | 18.5 | " |
| | | " | 116 | " | | | | 2.7 V | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 117 | " | | | | | | | " | OUT | GND | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 118 | " | | | | | | | " | | | | OUT | GND | 2.7 V | " | " | 4Y | " | " | " |
| | $t_{PLH3}$ | " | 119 | " | 2.7 V | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 18 | " |
| | | " | 120 | " | | | | 2.7 V | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 121 | " | | | | | | | " | OUT | GND | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 122 | " | | | | | | | " | | | | OUT | GND | 2.7 V | " | " | 4Y | " | " | " |
| 11 | Same tests, terminal conditions, and limits as for subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

1/ For circuit B, $I_{OS(max)}$ = -110 mA.

2/ A = 2.4 V; B = 0.4 V.

3/ H ≥ 1.5 V; L ≤ 1.5 V.

4/ Only a summary of attributes is required.

5/ Case 2 pins not designated are NC.

6/ For circuit B, 0.1/ -4 mA.

7/ For circuit B, 0.1/ -2 mA.





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TABLE III. Group A inspection for device type 04
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | G | $V_{CC}$ | | | | |
| 1 | $V_{OH}$ | 3006 | 1 | | | | -1 mA | | | | GND | | | | | | | 2.0 V | 4.5 V | 1Y | 2.5 | | V |
| $T_C$ = +25°C | | " | 2 | | | | | | | -1 mA | " | | | | | | | " | " | 2Y | " | | " |
| | | " | 3 | | | | | | | | " | -1 mA | | | | | | " | " | 3Y | " | | " |
| | | " | 4 | | | | | | | | " | | | | -1 mA | | | " | " | 4Y | " | | " |
| | $V_{OL}$ | 3007 | 5 | 2.0 V | | 2.0 V | 20 mA | | | | " | | | | | | | 0.8 V | " | 1Y | | 0.5 | " |
| | | " | 6 | " | | | | | 2.0 V | 20 mA | " | | | | | | | " | " | 2Y | | " | " |
| | | " | 7 | " | | | | | | | " | 20 mA | 2.0 V | | | | | " | " | 3Y | | " | " |
| | | " | 8 | " | | | | | | | " | | | | 20 mA | 2.0 V | | " | " | 4Y | | " | " |
| | $V_{IC}$ | | 9 | -18 mA | | | | | | | " | | | | | | | | " | S | | -1.2 | " |
| | | | 10 | | -18 mA | | | | | | " | | | | | | | | " | 1A | | " | " |
| | | | 11 | | | -18 mA | | | | | " | | | | | | | | " | 1B | | " | " |
| | | | 12 | | | | | -18 mA | | | " | | | | | | | | " | 2A | | " | " |
| | | | 13 | | | | | | -18 mA | | " | | | | | | | | " | 2B | | " | " |
| | | | 14 | | | | | | | | " | | -18 mA | | | | | | " | 3B | | " | " |
| | | | 15 | | | | | | | | " | | | -18 mA | | | | | " | 3A | | " | " |
| | | | 16 | | | | | | | | " | | | | | -18 mA | | | " | 4B | | " | " |
| | | | 17 | | | | | | | | " | | | | | | -18 mA | | " | 4A | | " | " |
| | | | 18 | | | | | | | | " | | | | | | | -18 mA | " | G | | " | " |
| | $I_{IL}$ | 3009 | 19 | 5.5 V | | | | | | | " | | | | | | | 0.5 V | 5.5 V | G | -2 6/ | -4 6/ | mA |
| | | " | 20 | 0.5 V | | | | | | | " | | | | | | | 5.5 V | " | S | -2 6/ | -4 6/ | " |
| | | " | 21 | GND | 0.5 V | | | | | | " | | | | | | | GND | " | 1A | -1 7/ | -2 7/ | " |
| | | " | 22 | 5.5 V | | 0.5 V | | | | | " | | | | | | | " | " | 1B | " " | " " | " |
| | | " | 23 | GND | | | | 0.5 V | | | " | | | | | | | " | " | 2A | " " | " " | " |
| | | " | 24 | 5.5 V | | | | | 0.5 V | | " | | | | | | | " | " | 2B | " " | " " | " |
| | | " | 25 | 5.5 V | | | | | | | " | | 0.5 V | | | | | " | " | 3B | " " | " " | " |
| | | " | 26 | GND | | | | | | | " | | | 0.5 V | | | | " | " | 3A | " " | " " | " |
| | | " | 27 | 5.5 V | | | | | | | " | | | | | 0.5 V | | " | " | 4B | " " | " " | " |
| | | " | 28 | GND | | | | | | | " | | | | | | 0.5 V | " | " | 4A | " " | " " | " |
| | $I_{IH1}$ | 3010 | 29 | GND | | | | | | | " | | | | | | | 2.7 V | " | G | | 100 | µA |
| | | " | 30 | 2.7 V | | | | | | | " | | | | | | | GND | " | S | | 100 | " |
| | | " | 31 | | 2.7 V | | | | | | " | | | | | | | 5.5 V | " | 1A | | 50 | " |
| | | " | 32 | | | 2.7 V | | | | | " | | | | | | | " | " | 1B | | " | " |
| | | " | 33 | | | | | 2.7 V | | | " | | | | | | | " | " | 2A | | " | " |
| | | " | 34 | | | | | | 2.7 V | | " | | | | | | | " | " | 2B | | " | " |
| | | " | 35 | | | | | | | | " | | 2.7 V | | | | | " | " | 3B | | " | " |
| | | " | 36 | | | | | | | | " | | | 2.7 V | | | | " | " | 3A | | " | " |
| | | " | 37 | | | | | | | | " | | | | | 2.7 V | | " | " | 4B | | " | " |
| | | " | 38 | | | | | | | | " | | | | | | 2.7 V | " | " | 4A | | " | " |

See footnotes at end of device type 04





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TABLE III. Group A inspection for device type 04 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | G | $V_{CC}$ | | | | |
| 1 | $I_{IH2}$ | 3010 | 39 | GND | | | | | | | GND | | | | | | | 5.5 V | 5.5 V | G | | 1.0 | mA |
| $T_C$ = +25°C | | " | 40 | 5.5 V | | | | | | | " | | | | | | | GND | " | S | | " | " |
| | | " | 41 | | 5.5 V | | | | | | " | | | | | | | 5.5 V | " | 1A | | " | " |
| | | " | 42 | | | 5.5 V | | | | | " | | | | | | | " | " | 1B | | " | " |
| | | " | 43 | | | | | 5.5 V | | | " | | | | | | | " | " | 2A | | " | " |
| | | " | 44 | | | | | | 5.5 V | | " | | | | | | | " | " | 2B | | " | " |
| | | " | 45 | | | | | | | | " | | 5.5 V | | | | | " | " | 3B | | " | " |
| | | " | 46 | | | | | | | | " | | | 5.5 V | | | | " | " | 3A | | " | " |
| | | " | 47 | | | | | | | | " | | | | 5.5 V | | | " | " | 4B | | " | " |
| | | " | 48 | | | | | | | | " | | | | | 5.5 V | | " | " | 4A | | " | " |
| | $I_{OS}$ | 3011 | 49 | | | | GND | | | | " | | | | | | | " | " | 1Y | -40 | -100 1/ | " |
| | | " | 50 | | | | | | | GND | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 51 | | | | | | | | " | GND | | | | | | " | " | 3Y | " | " | " |
| | | " | 52 | | | | | | | | " | | | | GND | | | " | " | 4Y | " | " | " |
| | $I_{CC}$ | 3005 | 53 | 5.5 V | GND | GND | | GND | GND | | " | | GND | GND | | GND | GND | " | " | $V_{CC}$ | | 61 | " |
| | $I_{CEX}$ | | 54 | " | " | " | 5.5 V | " | " | | " | | " | " | | " | " | " | " | 1Y | | 250 | µA |
| | | | 55 | " | " | " | | " | " | 5.5 V | " | | " | " | | " | " | " | " | 2Y | | " | " |
| | | | 56 | " | " | " | | " | " | | " | 5.5 V | " | " | | " | " | " | " | 3Y | | " | " |
| | | | 57 | " | " | " | | " | " | | " | | " | " | 5.5 V | " | " | " | " | 4Y | | " | " |
| 2 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. $V_{IL}$ = 0.7 V, $V_{OL(max)}$ = 0.45 V. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 7 | Truth | 3014 | 58 | B 2/ | A 2/ | B 2/ | L | A 2/ | B 2/ | L | GND | L | B 2/ | A 2/ | L | B 2/ | A 2/ | B | 4.5 V | See 3/ | | | |
| $T_C$ = +25°C | table | " | 59 | A | B | A | L | | | | " | | | | | | | " | " | " | | | |
| | test | " | 60 | A | A | B | H | | | | " | | | | | | | " | " | " | | | |
| | | " | 61 | B | A | B | L | | | | " | | | | | | | " | " | " | | | |
| | | " | 62 | B | B | A | H | | | | " | | | | | | | " | " | " | | | |
| | | " | 63 | A | | | | B | A | L | " | | | | | | | " | " | " | | | |
| | | " | 64 | A | | | | A | B | H | " | | | | | | | " | " | " | | | |
| | | " | 65 | B | | | | A | B | L | " | | | | | | | " | " | " | | | |
| | | " | 66 | B | | | | B | A | H | " | | | | | | | " | " | " | | | |
| | | " | 67 | A | | | | | | | " | L | A | B | | | | " | " | " | | | |
| | | " | 68 | A | | | | | | | " | H | B | A | | | | " | " | " | | | |
| | | " | 69 | B | | | | | | | " | L | B | A | | | | " | " | " | | | |
| | | " | 70 | B | | | | | | | " | H | A | B | | | | " | " | " | | | |
| | | " | 71 | A | | | | | | | " | | | | L | A | B | " | " | " | | | |
| | | " | 72 | A | | | | | | | " | | | | H | B | A | " | " | " | | | |
| | | " | 73 | B | | | | | | | " | | | | L | B | A | " | " | " | | | |
| | | " | 74 | B | | | | | | | " | | | | H | A | B | " | " | " | | | |

See footnotes at end of device type 04





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TABLE III. Group A inspection for device type 04 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | G | $V_{CC}$ | | | | |
| 8 4/ | Same tests, terminal conditions, and limits as for subgroup 7, except $T_C$ = +125°C and -55°C. | | | | | | | | | | | | | | | | | | | | | | |
| 9 | $t_{PHL1}$ | 3003 | 75 | IN | GND | 2.7 V | OUT | | | | GND | | | | | | | GND | 5.0 V | 1Y | 2.0 | 14 | ns |
| $T_C$ = +25°C | | Fig. 6 | 76 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 77 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 78 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PLH1}$ | " | 79 | " | GND | 2.7 V | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 80 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 81 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 82 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PHL2}$ | " | 83 | GND | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 8.0 | " |
| | | " | 84 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 85 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 86 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PLH2}$ | " | 87 | " | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 88 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 89 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 90 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PHL3}$ | " | 91 | " | 2.7 V | GND | OUT | | | | " | | | | | | | IN | " | 1Y | " | 14 | " |
| | | " | 92 | " | | | | 2.7 V | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 93 | " | | | | | | | " | OUT | GND | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 94 | " | | | | | | | " | | | | OUT | GND | 2.7 V | " | " | 4Y | " | " | " |
| | $t_{PLH3}$ | " | 95 | " | 2.7 V | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 13.5 | " |
| | | " | 96 | " | | | | 2.7 V | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 97 | " | | | | | | | " | OUT | GND | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 98 | " | | | | | | | " | | | | OUT | GND | 2.7 V | " | " | 4Y | " | " | " |
| 10 | $t_{PHL1}$ | " | 99 | IN | GND | 2.7 V | OUT | | | | " | | | | | | | GND | " | 1Y | " | 18.5 | " |
| $T_C$ = +125°C | | " | 100 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 101 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 102 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |

See footnotes at end of device type 04.





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TABLE III. Group A inspection for device type 04 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | G | $V_{CC}$ | | | | |
| 10 | $t_{PLH1}$ | 3003 | 103 | IN | GND | 2.7 V | OUT | | | | GND | | | | | | | GND | 5.0 V | 1Y | 2.0 | 18.5 | ns |
| $T_C$ = +125°C | | Fig. 6 | 104 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 105 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 106 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PHL2}$ | " | 107 | GND | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 11 | " |
| | | " | 108 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 109 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 110 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PLH2}$ | " | 111 | " | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 112 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 113 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 114 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PHL3}$ | " | 115 | " | 2.7 V | GND | OUT | | | | " | | | | | | | IN | " | 1Y | " | 18.5 | " |
| | | " | 116 | " | | | | 2.7 V | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 117 | " | | | | | | | " | OUT | GND | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 118 | " | | | | | | | " | | | | OUT | GND | 2.7 V | " | " | 4Y | " | " | " |
| | $t_{PLH3}$ | " | 119 | " | 2.7 V | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 18 | " |
| | | " | 120 | " | | | | 2.7 V | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 121 | " | | | | | | | " | OUT | GND | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 122 | " | | | | | | | " | | | | OUT | GND | 2.7 V | " | " | 4Y | " | " | " |
| 11 | Same tests, terminal conditions, and limits as for subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

1/ For circuit B, $I_{OS(max)}$ = -110 mA.

2/ A = 2.4 V; B = 0.4 V.

3/ H ≥ 1.5 V; L ≤ 1.5 V.

4/ Only a summary of attributes is required.

5/ Case 2 pins not designated are NC.

6/ For circuit B, 0.1/ -4 mA.

7/ For circuit B, 0.1/ -2 mA.



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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | D3 | D2 | D1 | D0 | Y | W | ST | GND | C | B | A | D7 | D6 | D5 | D4 | $V_{CC}$ | | | | |
| 1<br>$T_C$ = +25°C | $V_{OL}$ | 3007 | 1 | | | | 2.0 V | | 20 mA | 0.8 V | GND | 0.8 V | 0.8 V | 0.8 V | | | | | 4.5 V | W | | 0.5 | V |
| | | 3007 | 2 | | | | | 20 mA | | " | " | 2.0 V | 2.0 V | 2.0 V | 0.8 V | | | | " | Y | | 0.5 | " |
| | $V_{OH}$ | 3006 | 3 | | | | | | -2 mA | " | " | 2.0 V | 2.0 V | 2.0 V | 0.8 V | | | | " | W | 2.4 | | " |
| | | 3006 | 4 | | | | 2.0 V | -2 mA | | " | " | 0.8 V | 0.8 V | 0.8 V | | | | | " | Y | 2.4 | | " |
| | $I_{OFF1}$ | | 5 | | | | | | 2.7 V | 2.0 V | " | | | | | | | | 5.5 V | W | | 50 | µA |
| | | | 6 | | | | | 2.7 V | | " | " | | | | | | | | " | Y | | 50 | " |
| | $I_{OFF2}$ | | 7 | | | | | | 0.5 V | " | " | | | | | | | | " | W | | -50 | " |
| | | | 8 | | | | | 0.5 V | | " | " | | | | | | | | " | Y | | -50 | " |
| | $I_{OS}$ | 3011 | 9 | | | | | | GND | GND | " | 5.5 V | 5.5 V | 5.5 V | GND | | | | " | W | -40 | -100 1/ | mA |
| | | 3011 | 10 | | | | 5.5 V | GND | | GND | " | GND | GND | GND | | | | | " | Y | -40 | -100 1/ | " |
| | $I_{IH1}$ | 3010 | 11 | 2.7 V | | | | | | 5.5 V | " | 5.5 V | " | GND | | | | | " | D3 | | 50 | µA |
| | | " | 12 | | 2.7 V | | | | | " | " | " | " | 5.5 V | | | | | " | D2 | | " | " |
| | | " | 13 | | | 2.7 V | | | | " | " | " | 5.5 V | GND | | | | | " | D1 | | " | " |
| | | " | 14 | | | | 2.7 V | | | " | " | " | 5.5 V | 5.5 V | | | | | " | D0 | | " | " |
| | | " | 15 | | | | | | | 2.7 V | " | | | | | | | | " | ST | | " | " |
| | | " | 16 | | | | | | | | " | 2.7 V | | | | | | | " | C | | " | " |
| | | " | 17 | | | | | | | | " | | 2.7 V | | | | | | " | B | | " | " |
| | | " | 18 | | | | | | | | " | | | 2.7 V | | | | | " | A | | " | " |
| | | " | 19 | | | | | | | 5.5 V | " | GND | GND | GND | 2.7 V | | | | " | D7 | | " | " |
| | | " | 20 | | | | | | | " | " | " | GND | 5.5 V | | 2.7 V | | | " | D6 | | " | " |
| | | " | 21 | | | | | | | " | " | " | 5.5 V | GND | | | 2.7 V | | " | D5 | | " | " |
| | | " | 22 | | | | | | | " | " | " | 5.5 V | 5.5 V | | | | 2.7 V | " | D4 | | " | " |
| | $I_{IH2}$ | " | 23 | 5.5 V | | | | | | " | " | 5.5 V | GND | GND | | | | | " | D3 | | 1.0 | mA |
| | | " | 24 | | 5.5 V | | | | | " | " | " | GND | 5.5 V | | | | | " | D2 | | " | " |
| | | " | 25 | | | 5.5 V | | | | " | " | " | 5.5 V | GND | | | | | " | D1 | | " | " |
| | | " | 26 | | | | 5.5 V | | | " | " | " | 5.5 V | 5.5 V | | | | | " | D0 | | " | " |
| | | " | 27 | | | | | | | " | " | | | | | | | | " | ST | | " | " |
| | | " | 28 | | | | | | | | " | 5.5 V | | | | | | | " | C | | " | " |
| | | | 29 | | | | | | | | " | | 5.5 V | | | | | | " | B | | " | " |
| | | " | 30 | | | | | | | | " | | | 5.5 V | | | | | " | A | | " | " |
| | | " | 31 | | | | | | | 5.5 V | " | GND | GND | GND | 5.5 V | | | | " | D7 | | " | " |
| | | " | 32 | | | | | | | " | " | " | GND | 5.5 V | | 5.5 V | | | " | D6 | | " | " |
| | | " | 33 | | | | | | | " | " | " | 5.5 V | GND | | | 5.5 V | | " | D5 | | " | " |
| | | " | 34 | | | | | | | " | " | " | " | 5.5 V | | | | 5.5 V | " | D4 | | " | " |

See footnotes at end of device type 05.



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TABLE III. Group A inspection for device type 05 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | D3 | D2 | D1 | D0 | Y | W | ST | GND | C | B | A | D7 | D6 | D5 | D4 | $V_{CC}$ | | | | |
| 1 $T_C$ = +25°C | $I_{IL}$ | 3009 | 35 | 0.5 V | | | | | | GND | GND | GND | 5.5 V | 5.5 V | | | | | 5.5 V | D3 | -1 | -2 | mA |
| | | " | 36 | | 0.5 V | | | | | " | " | " | " | GND | | | | | " | D2 | " | " | " |
| | | " | 37 | | | 0.5 V | | | | " | " | " | GND | 5.5 V | | | | | " | D1 | " | " | " |
| | | " | 38 | | | | 0.5 V | | | " | " | " | GND | GND | | | | | " | D0 | " | " | " |
| | | " | 39 | | | | | | | 0.5 V | " | | | | | | | | " | ST | " | " | " |
| | | " | 40 | | | | | | | | " | 0.5 V | | | | | | | " | C | " | " | " |
| | | " | 41 | | | | | | | | " | | 0.5 V | | | | | | " | B | " | " | " |
| | | " | 42 | | | | | | | | " | | | 0.5 V | | | | | " | A | " | " | " |
| | | " | 43 | | | | | | | GND | " | 5.5 V | 5.5 V | 5.5 V | 0.5 V | | | | " | D7 | " | " | " |
| | | " | 44 | | | | | | | " | " | " | 5.5 V | GND | | 0.5 V | | | " | D6 | " | " | " |
| | | " | 45 | | | | | | | " | " | " | GND | 5.5 V | | | 0.5 V | | " | D5 | " | " | " |
| | | " | 46 | | | | | | | " | " | " | GND | GND | | | | 0.5 V | " | D4 | " | " | " |
| | $I_{CCO}$ | 3005 | 47 | 5.5 V | 5.5 V | 5.5 V | 5.5 V | | | 5.5 V | " | " | 5.5 V | 5.5 V | 5.5 V | 5.5 V | 5.5 V | 5.5 V | " | $V_{CC}$ | | 85 | " |
| | $I_{CEX}$ | | 48 | | | | | | 5.5 V | | " | " | 5.5 V | 5.5 V | GND | | | | " | W | | 250 | µA |
| | | | 49 | | | | 5.5 V | 5.5 V | | GND | " | GND | GND | GND | | | | | " | Y | | 250 | µA |
| | $V_{IC}$ | | 50 | -18 mA | | | | | | | " | | | | | | | | 4.5 V | D3 | | -1.2 | V |
| | | | 51 | | -18 mA | | | | | | " | | | | | | | | " | D2 | | " | " |
| | | | 52 | | | -18 mA | | | | | " | | | | | | | | " | D1 | | " | " |
| | | | 53 | | | | -18 mA | | | | " | | | | | | | | " | D0 | | " | " |
| | | | 54 | | | | | | | -18 mA | " | | | | | | | | " | ST | | " | " |
| | | | 55 | | | | | | | | " | -18 mA | | | | | | | " | C | | " | " |
| | | | 56 | | | | | | | | " | | -18 mA | | | | | | " | B | | " | " |
| | | | 57 | | | | | | | | " | | | -18 mA | | | | | " | A | | " | " |
| | | | 58 | | | | | | | | " | | | | -18 mA | | | | " | D7 | | " | " |
| | | | 59 | | | | | | | | " | | | | | -18 mA | | | " | D6 | | " | " |
| | | | 60 | | | | | | | | " | | | | | | -18 mA | | " | D5 | | " | " |
| | | | 61 | | | | | | | | " | | | | | | | -18 mA | " | D4 | | " | " |
| 2 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. $V_{IL}$ = 0.7 V, $V_{OL(max)}$ = 0.45 V. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 7 $T_C$ = +25°C | Truth table test | 3014 | 62 | A 2/ | A 2/ | A 2/ | B 2/ | L | H | B 2/ | GND | B 2/ | B 2/ | B 2/ | A 2/ | A 2/ | A 2/ | A 2/ | 4.5 V | 3/ | | | |
| | | " | 63 | B | B | B | A | H | L | " | " | " | " | B | B | B | B | B | " | " | | | |
| | | " | 64 | A | A | B | A | L | H | " | " | " | " | A | A | A | A | A | " | " | | | |
| | | " | 65 | B | B | A | B | H | L | " | " | " | " | A | B | B | B | B | " | " | | | |
| | | " | 66 | A | B | A | A | L | H | " | " | " | A | B | A | A | A | A | " | " | | | |
| | | " | 67 | B | A | B | B | H | L | " | " | " | " | B | B | B | B | B | " | " | | | |
| | | " | 68 | B | A | A | A | L | H | " | " | " | " | A | A | A | A | A | " | " | | | |
| | | " | 69 | A | B | B | B | H | L | " | " | " | " | A | B | B | B | B | " | " | | | |
| | | " | 70 | A | A | A | A | L | H | " | " | A | B | B | A | A | A | B | " | " | | | |
| | | " | 71 | B | B | B | B | H | L | " | " | " | " | B | B | B | B | A | " | " | | | |
| | | " | 72 | A | A | A | A | L | H | " | " | " | " | A | A | A | B | A | " | " | | | |
| | | " | 73 | B | B | B | B | H | L | " | " | " | " | A | B | B | A | B | " | " | | | |
| | | " | 74 | A | A | A | A | L | H | " | " | " | A | B | A | B | A | A | " | " | | | |
| | | " | 75 | B | B | B | B | H | L | " | " | " | " | B | B | A | B | B | " | " | | | |
| | | " | 76 | A | A | A | A | L | H | " | " | " | " | A | B | A | A | A | " | " | | | |
| | | " | 77 | B | B | B | B | H | L | " | " | " | " | A | A | B | B | B | " | " | | | |

See footnotes at end of device type 05.





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TABLE III. Group A inspection for device type 05 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | D3 | D2 | D1 | D0 | Y | W | ST | GND | C | B | A | D7 | D6 | D5 | D4 | $V_{CC}$ | | | | |
| 8 | Repeat subgroup 7 at $T_C$ = +125°C and -55°C. | | | | | | | | | | | | | | | | | | | | | | |
| 9 $T_C$ = +25°C | $t_{PHL1}$ | 3003 Fig. 8 | 78 | | 2.7 V | 2.7 V | GND | | OUT | GND | GND | GND | GND | IN | | | | | 5.0 V | W | 2.0 | 15.5 | ns |
| | | | 79 | | | | " | | " | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | | 80 | | | | " | | " | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PLH1}$ | " | 81 | | | 2.7 V | " | | " | " | " | GND | GND | IN | | | | | " | " | " | 17.0 | " |
| | | " | 82 | | 2.7 V | | " | | " | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 83 | | | | " | | " | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PHL2}$ | " | 84 | | | 2.7 V | " | OUT | | " | " | GND | GND | IN | | | | | " | Y | " | 21.5 | " |
| | | " | 85 | | 2.7 V | | " | " | | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 86 | | | | " | " | | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PLH2}$ | " | 87 | | | 2.7 V | " | " | | " | " | GND | GND | IN | | | | | " | " | " | 20.0 | " |
| | | " | 88 | | 2.7 V | | " | " | | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 89 | | | | " | " | | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PHL3}$ | " | 90 | | | IN | IN | | OUT | " | " | GND | GND | GND | | | | | " | W | " | 9.0 | " |
| | | " | 91 | | | IN | | | " | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 92 | | IN | | | | " | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 93 | IN | | | | | " | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 94 | | | | | | " | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 95 | | | | | | " | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 96 | | | | | | " | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 97 | | | | | | " | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| | $t_{PLH3}$ | " | 98 | | | IN | IN | | " | " | " | GND | GND | GND | | | | | " | " | " | " | " |
| | | " | 99 | | | IN | | | " | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 100 | | IN | | | | " | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 101 | IN | | | | | " | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 102 | | | | | | " | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 103 | | | | | | " | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 104 | | | | | | " | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 105 | | | | | | " | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| | $t_{PHI\ 4}$ | " | 106 | | | IN | IN | OUT | | " | " | GND | GND | GND | | | | | " | Y | " | 14.0 | " |
| | | " | 107 | | | IN | | " | | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 108 | | IN | | | " | | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 109 | IN | | | | " | | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 110 | | | | | " | | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 111 | | | | | " | | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 112 | | | | | " | | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 113 | | | | | " | | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |

See footnotes at end of device type 05



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TABLE III. Group A inspection for device type 05 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | D3 | D2 | D1 | D0 | Y | W | ST | GND | C | B | A | D7 | D6 | D5 | D4 | $V_{CC}$ | | | | |
| 9 | $t_{PLH4}$ | 3003 | 114 | | | | IN | OUT | | GND | GND | GND | GND | GND | | | | | 5.0 V | Y | 2.0 | 14.0 | ns |
| $T_C$ = +25°C | | Fig. 8 | 115 | | | IN | | " | | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 116 | | IN | | | " | | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 117 | IN | | | | " | | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 118 | | | | | " | | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 119 | | | | | " | | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 120 | | | | | " | | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 121 | | | | | " | | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| | $t_{ZH}$ | " | 122 | | | | GND | | OUT | IN | " | GND | GND | GND | | | | | " | W | " | 19.5 | " |
| | $t_{ZL}$ | " | 123 | | | | 2.7 V | | " | " | " | " | " | " | | | | | " | " | " | 21 | " |
| | $t_{HZ}$ | " | 124 | | | | GND | | " | " | " | " | " | " | | | | | " | " | " | 21 | " |
| | $t_{LZ}$ | " | 125 | | | | 2.7 V | | " | " | " | " | " | " | | | | | " | " | " | 17 | " |
| | $t_{ZH}$ | " | 126 | | | | 2.7 V | OUT | | " | " | " | " | " | | | | | " | Y | " | 19.5 | " |
| | $t_{ZL}$ | " | 127 | | | | GND | " | | " | " | " | " | " | | | | | " | " | " | 21 | " |
| | $t_{HZ}$ | " | 128 | | | | 2.7 V | " | | " | " | " | " | " | | | | | " | " | " | 21 | " |
| | $t_{LZ}$ | " | 129 | | | | GND | " | | " | " | " | " | " | | | | | " | " | " | 17 | " |
| 10 | $t_{PHL1}$ | " | 130 | | | 2.7 V | GND | | OUT | GND | " | GND | GND | IN | | | | | " | W | " | 20 | " |
| $T_C$ = +125°C | | " | 131 | | 2.7 V | | | | " | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 132 | | | | | | " | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PLH1}$ | " | 133 | | | 2.7 V | | | " | " | " | GND | GND | IN | | | | | " | " | " | 22 | " |
| | | " | 134 | | 2.7 V | | | | " | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 135 | | | | | | " | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PHL2}$ | " | 136 | | | 2.7 V | | OUT | | " | " | GND | GND | IN | | | | | " | Y | " | 28 | " |
| | | " | 137 | | 2.7 V | | | " | | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 138 | | | | | " | | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |
| | $t_{PLH2}$ | " | 139 | | | 2.7 V | | " | | " | " | GND | GND | IN | | | | | " | " | " | 26 | " |
| | | " | 140 | | 2.7 V | | | " | | " | " | GND | IN | GND | | | | | " | " | " | " | " |
| | | " | 141 | | | | GND | " | | " | " | IN | GND | GND | | | | 2.7 V | " | " | " | " | " |

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TABLE III. Group A inspection for device type 05 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | D3 | D2 | D1 | D0 | Y | W | ST | GND | C | B | A | D7 | D6 | D5 | D4 | $V_{CC}$ | | | | |
| 10<br>$T_C$ = +125°C | $t_{PHL3}$ | 3003<br>Fig. 8 | 142 | | | | IN | | OUT | GND | GND | GND | GND | GND | | | | | 5.0 V | W | 2.0 | 11.5 | ns |
| | | | 143 | | | IN | | | " | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | | 144 | | IN | | | | " | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | | 145 | IN | | | | | " | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | | 146 | | | | | | " | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | | 147 | | | | | | " | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | | 148 | | | | | | " | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | | 149 | | | | | | " | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| | $t_{PLH3}$ | " | 150 | | | | IN | | " | " | " | GND | GND | GND | | | | | " | " | " | " | " |
| | | " | 151 | | | IN | | | " | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 152 | | IN | | | | " | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 153 | IN | | | | | " | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 154 | | | | | | " | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 155 | | | | | | " | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 156 | | | | | | " | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 157 | | | | | | " | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| | $t_{PHL4}$ | " | 158 | | | | IN | OUT | | " | " | GND | GND | GND | | | | | " | Y | " | 18 | " |
| | | " | 159 | | | IN | | " | | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 160 | | IN | | | " | | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 161 | IN | | | | " | | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 162 | | | | | " | | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 163 | | | | | " | | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 164 | | | | | " | | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 165 | | | | | " | | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| | $t_{PLH4}$ | " | 166 | | | | IN | " | | " | " | GND | GND | GND | | | | | " | " | " | " | " |
| | | " | 167 | | | IN | | " | | " | " | " | GND | 2.7 V | | | | | " | " | " | " | " |
| | | " | 168 | | IN | | | " | | " | " | " | 2.7 V | GND | | | | | " | " | " | " | " |
| | | " | 169 | IN | | | | " | | " | " | " | 2.7 V | 2.7 V | | | | | " | " | " | " | " |
| | | " | 170 | | | | | " | | " | " | 2.7 V | GND | GND | | | | IN | " | " | " | " | " |
| | | " | 171 | | | | | " | | " | " | " | GND | 2.7 V | | | IN | | " | " | " | " | " |
| | | " | 172 | | | | | " | | " | " | " | 2.7 V | GND | | IN | | | " | " | " | " | " |
| | | " | 173 | | | | | " | | " | " | " | 2.7 V | 2.7 V | IN | | | | " | " | " | " | " |
| | $t_{ZH1}$ | " | 174 | | | | GND | | OUT | IN | " | GND | GND | GND | | | | | " | W | " | 25.5 | " |
| | $t_{ZL1}$ | " | 175 | | | | 2.7 V | | " | " | " | " | " | " | | | | | " | " | " | 27.5 | " |
| | $t_{HZ2}$ | " | 176 | | | | GND | | " | " | " | " | " | " | | | | | " | " | " | 24 | " |
| | $t_{LZ2}$ | " | 177 | | | | 2.7 V | | " | " | " | " | " | " | | | | | " | " | " | 22 | " |
| | $t_{ZH3}$ | " | 178 | | | | 2.7 V | OUT | | " | " | " | " | " | | | | | " | Y | " | 25.5 | " |

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TABLE III. Group A inspection for device type 05 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | D3 | D2 | D1 | D0 | Y | W | ST | GND | C | B | A | D7 | D6 | D5 | D4 | $V_{CC}$ | | | | |
| 10 | $t_{ZL3}$ | 3003 Fig. 8 | 179 | | | | GND | OUT | | IN | GND | GND | GND | GND | | | | | 5.0 V | Y | 2.0 | 27.5 | ns |
| $T_C$ = +125°C | $t_{HZ4}$ | " | 180 | | | | 2.7 V | " | | " | " | " | " | " | | | | | " | " | " | 24 | " |
| | $t_{LZ4}$ | " | 181 | | | | GND | " | | " | " | " | " | " | | | | | " | " | " | 22 | " |
| 11 | Same tests, terminal conditions, and limits as for subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

1/ For circuit B, $I_{OS(max)}$ = -110 mA.

2/ A = 2.4 V; B = 0.4 V.

3/ H ≥ 1.5 V; L ≤ 1.5 V.

4/ Only a summary of attributes is required.

5/ Case 2 pins not designated are NC.



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TABLE III. Group A inspection for device type 06
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 7/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | 0E | $V_{CC}$ | | | | |
| 1<br>$T_C$ = +25°C | $V_{OH}$ | 3006 | 1 | 2.0 V | | 2.0 V | -2 mA | | | | GND | | | | | | | 0.8 V | 4.5 V | 1Y | 2.4 | | V |
| | | " | 2 | " | | | | | 2.0 V | -2 mA | " | | | | | | | " | " | 2Y | " | | " |
| | | " | 3 | " | | | | | | | " | -2 mA | 2.0 V | | | | | " | " | 3Y | " | | " |
| | | " | 4 | " | | | | | | | " | | | | -2 mA | 2.0 V | | " | " | 4Y | " | | " |
| | $V_{OL}$ | 3007 | 5 | 0.8 V | 0.8 V | | 20 mA | | | | " | | | | | | | " | " | 1Y | | 0.5 | " |
| | | " | 6 | " | | | | 0.8 V | | 20 mA | " | | | | | | | " | " | 2Y | | " | " |
| | | " | 7 | " | | | | | | | " | 20 mA | | 0.8 V | | | | " | " | 3Y | | " | " |
| | | " | 8 | " | | | | | | | " | | | | 20 mA | | 0.8 V | " | " | 4Y | | " | " |
| | $I_{OFF1}$ | | 9 | GND | GND | | 2.7 V | | | | " | | | | | | | 2.0 V | 5.5 V | 1Y | | 50 | µA |
| | | | 10 | " | | | | GND | | 2.7 V | " | | | | | | | " | " | 2Y | | " | " |
| | | | 11 | " | | | | | | | " | 2.7 V | | GND | | | | " | " | 3Y | | " | " |
| | | | 12 | " | | | | | | | " | | | | 2.7 V | | GND | " | " | 4Y | | " | " |
| | $I_{OFF2}$ | | 13 | 5.5 V | | 5.5 V | 0.5 V | | | | " | | | | | | | " | " | 1Y | | -50 | " |
| | | | 14 | " | | | | | 5.5 V | 0.5 V | " | | | | | | | " | " | 2Y | | " | " |
| | | | 15 | " | | | | | | | " | 0.5 V | 5.5 V | | | | | " | " | 3Y | | " | " |
| | | | 16 | " | | | | | | | " | | | | 0.5 V | 5.5 V | | " | " | 4Y | | " | " |
| | $I_{CEX}$ | | 17 | 5.5 V | | 5.5 V | 5.5 V | | | | " | | | | | | | GND | " | 1Y | | 250 | " |
| | | | 18 | " | | | | | 5.5 V | 5.5 V | " | | | | | | | " | " | 2Y | | " | " |
| | | | 19 | " | | | | | | | " | 5.5 V | 5.5 V | | | | | " | " | 3Y | | " | " |
| | | | 20 | " | | | | | | | " | | | | 5.5 V | 5.5 V | | " | " | 4Y | | " | " |
| | $I_{IL}$ | 3009 | 21 | 0.5 V | | | | | | | " | | | | | | | | " | S | -2.0 8/ | -4.0 8/ | mA |
| | | " | 22 | GND | | | | | | | " | | | | | | | 0.5 V | " | 0E | -1.0 9/ | -2.0 9/ | " |
| | | " | 23 | GND | 0.5 V | | | | | | " | | | | | | | | " | 1A | " | " | " |
| | | " | 24 | 5.5 V | | 0.5 V | | | | | " | | | | | | | | " | 1B | " | " | " |
| | | " | 25 | GND | | | | 0.5 V | | | " | | | | | | | | " | 2A | " | " | " |
| | | " | 26 | 5.5 V | | | | | 0.5 V | | " | | | | | | | | " | 2B | " | " | " |
| | | " | 27 | 5.5 V | | | | | | | " | | 0.5 V | | | | | | " | 3B | " | " | " |
| | | " | 28 | GND | | | | | | | " | | | 0.5 V | | | | | " | 3A | " | " | " |
| | | " | 29 | 5.5 V | | | | | | | " | | | | | 0.5 V | | | " | 4B | " | " | " |
| | | " | 30 | GND | | | | | | | " | | | | | | 0.5 V | | " | 4A | " | " | " |
| | $I_{IH1}$ | 3010 | 31 | 2.7 V | | | | | | | " | | | | | | | | " | S | | 100 | µA |
| | | " | 32 | 5.5 V | | | | | | | " | | | | | | | 2.7 V | " | 0E | | 50 | " |
| | | " | 33 | 5.5 V | 2.7 V | | | | | | " | | | | | | | | " | 1A | | " | " |
| | | " | 34 | GND | | 2.7 V | | | | | " | | | | | | | | " | 1B | | " | " |
| | | " | 35 | 5.5 V | | | | 2.7 V | | | " | | | | | | | | " | 2A | | " | " |
| | | " | 36 | GND | | | | | 2.7 V | | " | | | | | | | | " | 2B | | " | " |
| | | " | 37 | GND | | | | | | | " | | 2.7 V | | | | | | " | 3B | | " | " |
| | | " | 38 | 5.5 V | | | | | | | " | | | 2.7 V | | | | | " | 3A | | " | " |
| | | " | 39 | GND | | | | | | | " | | | | | 2.7 V | | | " | 4B | | " | " |
| | | " | 40 | 5.5 V | | | | | | | " | | | | | | 2.7 V | | " | 4A | | " | " |

See footnotes at end of device type 06





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TABLE III. Group A inspection for device type 06 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 7/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | 0E | $V_{CC}$ | | | | |
| 1 | $I_{IH2}$ | 3010 | 41 | 5.5 V | | | | | | | GND | | | | | | | | 5.5 V | S | | 1.0 | mA |
| $T_C$ = +25°C | | " | 42 | 5.5 V | | | | | | | " | | | | | | | 5.5 V | " | 0E | | " | " |
| | | " | 43 | 5.5 V | 5.5 V | | | | | | " | | | | | | | | " | 1A | | " | " |
| | | " | 44 | GND | | 5.5 V | | | | | " | | | | | | | | " | 1B | | " | " |
| | | " | 45 | 5.5 V | | | | 5.5 V | | | " | | | | | | | | " | 2A | | " | " |
| | | " | 46 | GND | | | | | 5.5 V | | " | | | | | | | | " | 2B | | " | " |
| | | " | 47 | GND | | | | | | | " | | 5.5 V | | | | | | " | 3B | | " | " |
| | | " | 48 | 5.5 V | | | | | | | " | | | 5.5 V | | | | | " | 3A | | " | " |
| | | " | 49 | GND | | | | | | | " | | | | | 5.5 V | | | " | 4B | | " | " |
| | | " | 50 | 5.5 V | | | | | | | " | | | | | | 5.5 V | | " | 4A | | " | " |
| | $I_{OS}$ | 3011 | 51 | " | | 5.5 V | GND | | | | " | | | | | | | GND | " | 1Y | -40 | -100 1/ | " |
| | | " | 52 | " | | | | | 5.5 V | GND | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 53 | " | | | | | | | " | GND | 5.5 V | | | | | " | " | 3Y | " | " | " |
| | | " | 54 | " | | | | | | | " | | | | GND | 5.5 V | | " | " | 4Y | " | " | " |
| | $I_{CCH}$ | 3005 | 55 | " | GND | 5.5 V | | GND | 5.5 V | | " | | 5.5 V | GND | | 5.5 V | GND | " | " | $V_{CC}$ | | 68 | " |
| | $I_{CCL}$ | " | 56 | GND | GND | GND | | GND | GND | | " | | GND | GND | | GND | GND | " | " | $V_{CC}$ | | 93 | " |
| | $I_{CCO}$ | " | 57 | GND | GND | GND | | GND | GND | | " | | GND | GND | | GND | GND | 5.5 V | " | $V_{CC}$ | | 99 | " |
| | $V_{IC}$ | | 58 | -18 mA | | | | | | | " | | | | | | | | 4.5 V | S | | -1.2 | V |
| | | | 59 | | -18 mA | | | | | | " | | | | | | | | " | 1A | | " | " |
| | | | 60 | | | -18 mA | | | | | " | | | | | | | | " | 1B | | " | " |
| | | | 61 | | | | | -18 mA | | | " | | | | | | | | " | 2A | | " | " |
| | | | 62 | | | | | | -18 mA | | " | | | | | | | | " | 2B | | " | " |
| | | | 63 | | | | | | | | " | | -18 mA | | | | | | " | 3B | | " | " |
| | | | 64 | | | | | | | | " | | | -18 mA | | | | | " | 3A | | " | " |
| | | | 65 | | | | | | | | " | | | | | -18 mA | | | " | 4B | | " | " |
| | | | 66 | | | | | | | | " | | | | | | -18 mA | | " | 4A | | " | " |
| | | | 67 | | | | | | | | " | | | | | | | -18 mA | " | 0E | | " | " |
| 2 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. $V_{IL}$ = 0.7 V, $V_{OL(max)}$ = 0.45 V. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |

See footnotes at end of device type 06




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TABLE III. Group A inspection for device type 06 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 7/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | 0E | $V_{CC}$ | | | | |
| 7 | Truth | 3014 | 68 | A 2/ | A 2/ | B 2/ | L | | | | GND | | | | | | | B 2/ | 4.5 V | See 3/ | | | |
| $T_C$ = +25°C | table | " | 69 | A | B | A | H | | | | " | | | | | | | " | " | " | | | |
| | test | " | 70 | B | B | A | L | | | | " | | | | | | | " | " | " | | | |
| | | " | 71 | B | A | B | H | | | | " | | | | | | | " | " | " | | | |
| | | " | 72 | A | | | | A | B | L | " | | | | | | | " | " | " | | | |
| | | " | 73 | A | | | | B | A | H | " | | | | | | | " | " | " | | | |
| | | " | 74 | B | | | | B | A | L | " | | | | | | | " | " | " | | | |
| | | " | 75 | B | | | | A | B | H | " | | | | | | | " | " | " | | | |
| | | " | 76 | A | | | | | | | " | L | B | A | | | | " | " | " | | | |
| | | " | 77 | A | | | | | | | " | H | A | B | | | | " | " | " | | | |
| | | " | 78 | B | | | | | | | " | L | A | B | | | | " | " | " | | | |
| | | " | 79 | B | | | | | | | " | H | B | A | | | | " | " | " | | | |
| | | " | 80 | A | | | | | | | " | | | | L | B | A | " | " | " | | | |
| | | " | 81 | A | | | | | | | " | | | | H | A | B | " | " | " | | | |
| | | " | 82 | B | | | | | | | " | | | | L | A | B | " | " | " | | | |
| | | " | 83 | B | | | | | | | " | | | | H | B | A | " | " | " | | | |
| 8 | Same tests, terminal conditions, and limits as for subgroup 7, except $T_C$ = +125°C and -55°C. | | | | | | | | | | | | | | | | | | | | | | |
| 9 | $t_{PHL1}$ | 3003 | 84 | IN | GND | 2.7 V | OUT | | | | GND | | | | | | | GND | 5.0 V | 1Y | 2.0 | 17 | ns |
| $T_C$ = +25°C | | Fig. 9 | 85 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 86 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 87 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PLH1}$ | " | 88 | IN | GND | 2.7 V | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 89 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 90 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 91 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |

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TABLE III. Group A inspection for device type 06 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 7/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | 0E | $V_{CC}$ | | | | |
| 9 | $t_{PHL2}$ | 3003 | 92 | GND | IN | GND | OUT | | | | GND | | | | | | | GND | 5.0 V | 1Y | 2.0 | 8.0 | ns |
| $T_C$ = +25°C | | Fig. 9 | 93 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 94 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 95 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PLH2}$ | " | 96 | " | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 9.0 | " |
| | | " | 97 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 98 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 99 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{ZH}$ | " | 100 | 2.7 V | | 2.7 V | OUT | | | | " | | | | | | | IN | " | 1Y | " | 21.5 | " |
| | | " | 101 | " | | | | | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 102 | " | | | | | | | " | OUT | 2.7 V | | | | | " | " | 3Y | " | " | " |
| | | " | 103 | " | | | | | | | " | | | | OUT | 2.7 V | | " | " | 4Y | " | " | " |
| | $t_{ZL}$ | " | 104 | GND | GND | | OUT | | | | " | | | | | | | " | " | 1Y | " | 23 | " |
| | | " | 105 | " | | | | GND | | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 106 | " | | | | | | | " | OUT | | GND | | | | " | " | 3Y | " | " | " |
| | | " | 107 | " | | | | | | | " | | | | OUT | | GND | " | " | 4Y | " | " | " |
| | $t_{HZ}$ 5/ | " | 108 | 2.7 V | | 2.7 V | OUT | | | | " | | | | | | | " | " | 1Y | " | 11.5 | " |
| | | " | 109 | " | | | | | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 110 | " | | | | | | | " | OUT | 2.7 V | | | | | " | " | 3Y | " | " | " |
| | | " | 111 | " | | | | | | | " | | | | OUT | 2.7 V | | " | " | 4Y | " | " | " |
| | $t_{LZ}$ | " | 112 | GND | GND | | OUT | | | | " | | | | | | | " | " | 1Y | " | 17 | " |
| | | " | 113 | " | | | | GND | | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 114 | " | | | | | | | " | OUT | | GND | | | | " | " | 3Y | " | " | " |
| | | " | 115 | " | | | | | | | " | | | | OUT | | GND | " | " | 4Y | " | " | " |
| 10 | $t_{PHL1}$ | " | 116 | IN | GND | 2.7 V | OUT | | | | " | | | | | | | GND | " | 1Y | " | 22 | " |
| $T_C$ = +125°C | | " | 117 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 118 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 119 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PLH1}$ | " | 120 | " | GND | 2.7 V | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 121 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 122 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 123 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |

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TABLE III. Group A inspection for device type 06 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 7/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | 0E | $V_{CC}$ | | | | |
| 10 | $t_{PHL2}$ | 3003 | 124 | GND | IN | GND | OUT | | | | GND | | | | | | | GND | 5.0 V | 1Y | 2.0 | 11 | ns |
| $T_C$ = +125°C | | Fig. 9 | 125 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 126 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 127 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PLH2}$ | " | 128 | " | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 12 | " |
| | | " | 129 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 130 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 131 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{ZH}$ | " | 132 | 2.7 V | | 2.7 V | OUT | | | | " | | | | | | | IN | " | 1Y | " | 28 | " |
| | | " | 133 | " | | | | | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 134 | " | | | | | | | " | OUT | 2.7 V | | | | | " | " | 3Y | " | " | " |
| | | " | 135 | " | | | | | | | " | | | | OUT | 2.7 V | | " | " | 4Y | " | " | " |
| | $t_{ZL}$ | " | 136 | GND | GND | | OUT | | | | " | | | | | | | " | " | 1Y | " | 30 | " |
| | | " | 137 | " | | | | GND | | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 138 | " | | | | | | | " | OUT | | GND | | | | " | " | 3Y | " | " | " |
| | | " | 139 | " | | | | | | | " | | | | OUT | | GND | " | " | 4Y | " | " | " |
| | $t_{HZ}$ 6/ | " | 140 | 2.7 V | | 2.7 V | OUT | | | | " | | | | | | | " | " | 1Y | " | 15 | " |
| | | " | 141 | " | | | | | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 142 | " | | | | | | | " | OUT | 2.7 V | | | | | " | " | 3Y | " | " | " |
| | | " | 143 | " | | | | | | | " | | | | OUT | 2.7 V | | IN | " | 4Y | " | " | " |
| | $t_{LZ}$ | " | 144 | GND | GND | | OUT | | | | " | | | | | | | " | " | 1Y | " | 22 | " |
| | | " | 145 | " | | | | GND | | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 146 | " | | | | | | | " | OUT | | GND | | | | " | " | 3Y | " | " | " |
| | | " | 147 | " | | | | | | | " | | | | OUT | | GND | " | " | 4Y | " | " | " |
| 11 | Same tests, terminal conditions, and limits as for subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

1/ For circuit B, $I_{OS(max)}$ = -110 mA.

2/ A = 2.4 V; B = 0.4 V.

3/ H ≥ 1.5 V; L ≤ 1.5 V.

4/ Only a summary of attributes is required.

5/ $t_{HZ}$ maximum limit for circuit C is 22 ns.

6/ $t_{HZ}$ maximum limit for circuit C is 24 ns.

7/ Case 2 pins not designated are NC.

8/ $I_{IL}$ limits for circuit B shall be -0.005 mA min / -0.1 mA max.

9/ $I_{IL}$ limits for circuit B shall be -0.005 mA min / -0.05 mA max.



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TABLE III. Group A inspection for device type 07
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 7/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | 0E | $V_{CC}$ | | | | |
| 1<br>$T_C$ = +25°C | $V_{OH}$ | 3006 | 1 | 2.0 V | | 0.8 V | -2.0 mA | | 0.8 V | -2.0 mA | GND | | | | | | | 0.8 V | 4.5 V | 1Y | 2.4 | | V |
| | " | | 2 | " | | | | | | | " | | | | | | | " | " | 2Y | " | | " |
| | " | | 3 | " | | | | | | | " | -2.0 mA | 0.8 V | | | | | " | " | 3Y | " | | " |
| | " | | 4 | " | | | | | | | " | | | | -2.0 mA | 0.8 V | | " | " | 4Y | " | | " |
| | $V_{OL}$ | 3007 | 5 | 0.8 V | 2.0 V | | 20 mA | | | | " | | | | | | | " | " | 1Y | | 0.5 | " |
| | " | | 6 | " | | | | 2.0 V | | 20 mA | " | | | | | | | " | " | 2Y | | " | " |
| | " | | 7 | " | | | | | | | " | 20 mA | | 2.0 V | | | | " | " | 3Y | | " | " |
| | " | | 8 | " | | | | | | | " | | | | 20 mA | | 2.0 V | " | " | 4Y | | " | " |
| | $I_{OFF1}$ | | 9 | GND | 5.5 V | | 2.7 V | | | | " | | | | | | | 2.0 V | 5.5 V | 1Y | | 50 | µA |
| | | | 10 | " | | | | 5.5 V | | 2.7 V | " | | | | | | | " | " | 2Y | | " | " |
| | | | 11 | " | | | | | | | " | 2.7 V | | 5.5 V | | | | " | " | 3Y | | " | " |
| | | | 12 | " | | | | | | | " | | | | 2.7 V | | 5.5 V | " | " | 4Y | | " | " |
| | $I_{OFF2}$ | | 13 | 5.5 V | | GND | 0.5 V | | | | " | | | | | | | " | " | 1Y | | -50 | " |
| | | | 14 | " | | | | | GND | 0.5 V | " | | | | | | | " | " | 2Y | | " | " |
| | | | 15 | " | | | | | | | " | 0.5 V | GND | | | | | " | " | 3Y | | " | " |
| | | | 16 | " | | | | | | | " | | | | 0.5 V | GND | | " | " | 4Y | | " | " |
| | $I_{CEX}$ | | 17 | 5.5 V | | GND | 5.5 V | | | | " | | | | | | | GND | " | 1Y | | 250 | " |
| | | | 18 | " | | | | | GND | 5.5 V | " | | | | | | | " | " | 2Y | | " | " |
| | | | 19 | " | | | | | | | " | 5.5 V | GND | | | | | " | " | 3Y | | " | " |
| | | | 20 | " | | | | | | | " | | | | 5.5 V | GND | | " | " | 4Y | | " | " |
| | $I_{IL}$ | 3009 | 21 | 0.5 V | | | | | | | " | | | | | | | | " | S | -2.0 | -4.0 | mA |
| | " | | 22 | GND | | | | | | | " | | | | | | | 0.5 V | " | 0E | -1.0 | -2.0 | " |
| | " | | 23 | GND | 0.5 V | | | | | | " | | | | | | | | " | 1A | " | " | " |
| | " | | 24 | 5.5 V | | 0.5 V | | | | | " | | | | | | | | " | 1B | " | " | " |
| | " | | 25 | GND | | | | 0.5 V | | | " | | | | | | | | " | 2A | " | " | " |
| | " | | 26 | 5.5 V | | | | | 0.5 V | | " | | | | | | | | " | 2B | " | " | " |
| | " | | 27 | 5.5 V | | | | | | | " | | 0.5 V | | | | | | " | 3B | " | " | " |
| | " | | 28 | GND | | | | | | | " | | | 0.5 V | | | | | " | 3A | " | " | " |
| | " | | 29 | 5.5 V | | | | | | | " | | | | | 0.5 V | | | " | 4B | " | " | " |
| | " | | 30 | GND | | | | | | | " | | | | | | 0.5 V | | " | 4A | " | " | " |
| | $I_{IH1}$ | 3010 | 31 | 2.7 V | | | | | | | " | | | | | | | | " | S | | 100 | µA |
| | " | | 32 | 5.5 V | | | | | | | " | | | | | | | 2.7 V | " | 0E | | 50 | " |
| | " | | 33 | 5.5 V | 2.7 V | | | | | | " | | | | | | | | " | 1A | | " | " |
| | " | | 34 | GND | | 2.7 V | | | | | " | | | | | | | | " | 1B | | " | " |
| | " | | 35 | 5.5 V | | | | 2.7 V | | | " | | | | | | | | " | 2A | | " | " |
| | " | | 36 | GND | | | | | 2.7 V | | " | | | | | | | | " | 2B | | " | " |
| | " | | 37 | GND | | | | | | | " | | 2.7 V | | | | | | " | 3B | | " | " |
| | " | | 38 | 5.5 V | | | | | | | " | | | 2.7 V | | | | | " | 3A | | " | " |
| | " | | 39 | GND | | | | | | | " | | | | | 2.7 V | | | " | 4B | | " | " |
| | " | | 40 | 5.5 V | | | | | | | " | | | | | | 2.7 V | | " | 4A | | " | " |

See footnotes at end of device type 07



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TABLE III. Group A inspection for device type 07 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 7/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | 0E | $V_{CC}$ | | | | |
| 1<br>$T_C$ = +25°C | $I_{IH2}$ | 3010 | 41 | 5.5 V | | | | | | | GND | | | | | | | | 5.5 V | S | | 1.0 | mA |
| | | " | 42 | 5.5 V | | | | | | | " | | | | | | | 5.5 V | " | 0E | | " | " |
| | | " | 43 | 5.5 V | 5.5 V | | | | | | " | | | | | | | | " | 1A | | " | " |
| | | " | 44 | GND | | 5.5 V | | | | | " | | | | | | | | " | 1B | | " | " |
| | | " | 45 | 5.5 V | | | | 5.5 V | | | " | | | | | | | | " | 2A | | " | " |
| | | " | 46 | GND | | | | | 5.5 V | | " | | | | | | | | " | 2B | | " | " |
| | | " | 47 | GND | | | | | | | " | | 5.5 V | | | | | | " | 3B | | " | " |
| | | " | 48 | 5.5 V | | | | | | | " | | | 5.5 V | | | | | " | 3A | | " | " |
| | | " | 49 | GND | | | | | | | " | | | | | 5.5 V | | | " | 4B | | " | " |
| | | " | 50 | 5.5 V | | | | | | | " | | | | | | 5.5 V | | " | 4A | | " | " |
| | $I_{OS}$ | 3011 | 51 | 5.5 V | | GND | GND | | | | " | | | | | | | GND | " | 1Y | -40 | -100 1/ | " |
| | | " | 52 | " | | | | | GND | GND | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 53 | " | | | | | | | " | GND | GND | | | | | " | " | 3Y | " | " | " |
| | | " | 54 | " | | | | | | | " | | | | GND | GND | | " | " | 4Y | " | " | " |
| | $I_{CCH}$ | 3005 | 55 | GND | GND | GND | | GND | GND | | " | | GND | GND | | GND | GND | " | " | $V_{CC}$ | | 56 | " |
| | $I_{CCL}$ | " | 56 | 5.5 V | GND | 5.5 V | | GND | 5.5 V | | " | | 5.5 V | GND | | 5.5 V | GND | " | " | $V_{CC}$ | | 81 | " |
| | $I_{CCO}$ | " | 57 | GND | GND | GND | | GND | GND | | " | | GND | GND | | GND | GND | 5.5 V | " | $V_{CC}$ | | 87 | " |
| | $V_{IC}$ | | 58 | -18 mA | | | | | | | " | | | | | | | | 4.5 V | S | | -1.2 | V |
| | | | 59 | | -18 mA | | | | | | " | | | | | | | | " | 1A | | " | " |
| | | | 60 | | | -18 mA | | | | | " | | | | | | | | " | 1B | | " | " |
| | | | 61 | | | | | -18 mA | | | " | | | | | | | | " | 2A | | " | " |
| | | | 62 | | | | | | -18 mA | | " | | | | | | | | " | 2B | | " | " |
| | | | 63 | | | | | | | | " | | -18 mA | | | | | | " | 3B | | " | " |
| | | | 64 | | | | | | | | " | | | -18 mA | | | | | " | 3A | | " | " |
| | | | 65 | | | | | | | | " | | | | | -18 mA | | | " | 4B | | " | " |
| | | | 66 | | | | | | | | " | | | | | | -18 mA | | " | 4A | | " | " |
| | | | 67 | | | | | | | | " | | | | | | | -18 mA | " | 0E | | " | " |
| 2 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. $V_{IL}$ = 0.7 V, $V_{OL(max)}$ = 0.45 V. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |

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TABLE III. Group A inspection for device type 07 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 7/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | 0E | $V_{CC}$ | | | | |
| 7 4/ $T_C$ = +25°C | Truth table test | 3014 | 68 | A 2/ | A 2/ | B 2/ | H | | | | GND | | | | | | | B 2/ | 4.5 V | See 3/ | | | |
| | | " | 69 | A | B | A | L | | | | " | | | | | | | " | " | " | | | |
| | | " | 70 | B | B | A | H | | | | " | | | | | | | " | " | " | | | |
| | | " | 71 | B | A | B | L | | | | " | | | | | | | " | " | " | | | |
| | | " | 72 | A | | | | A | B | H | " | | | | | | | " | " | " | | | |
| | | " | 73 | A | | | | B | A | L | " | | | | | | | " | " | " | | | |
| | | " | 74 | B | | | | B | A | H | " | | | | | | | " | " | " | | | |
| | | " | 75 | B | | | | A | B | L | " | | | | | | | " | " | " | | | |
| | | " | 76 | A | | | | | | | " | H | B | A | | | | " | " | " | | | |
| | | " | 77 | A | | | | | | | " | L | A | B | | | | " | " | " | | | |
| | | " | 78 | B | | | | | | | " | H | A | B | | | | " | " | " | | | |
| | | " | 79 | B | | | | | | | " | L | B | A | | | | " | " | " | | | |
| | | " | 80 | A | | | | | | | " | | | | H | B | A | " | " | " | | | |
| | | " | 81 | A | | | | | | | " | | | | L | A | B | " | " | " | | | |
| | | " | 82 | B | | | | | | | " | | | | H | A | B | " | " | " | | | |
| | | | 83 | B | | | | | | | " | | | | L | B | A | " | " | " | | | |
| 8 4/ | Same tests, terminal conditions, and limits as for subgroup 7, except $T_C$ = +125°C and $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |
| 9 $T_C$ = +25°C | $t_{PHL1}$ | 3003 Fig. 10 | 84 | IN | GND | 2.7 V | OUT | | | | GND | | | | | | | GND | 5.0 V | 1Y | 2.0 | 14 | ns |
| | | | 85 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 86 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 87 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PLH1}$ | " | 88 | " | GND | 2.7 V | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 89 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 90 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 91 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PHL2}$ | " | 92 | GND | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 8.0 | " |
| | | " | 93 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 94 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 95 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PLH2}$ | " | 96 | " | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 97 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 98 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 99 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |

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TABLE III. Group A inspection for device type 07 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 7/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | 0E | $V_{CC}$ | | | | |
| 9 | $t_{ZH}$ | 3003 | 100 | 2.7 V | | GND | OUT | | | | GND | | | | | | | IN | 5.0 V | 1Y | 2.0 | 21.5 | ns |
| $T_C$ = +25°C | | Fig. 10 | 101 | " | | | | | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 102 | " | | | | | | | " | OUT | GND | | | | | " | " | 3Y | " | " | " |
| | | " | 103 | " | | | | | | | " | | | | OUT | GND | | " | " | 4Y | " | " | " |
| | $t_{ZL}$ | " | 104 | GND | 2.7 V | | OUT | | | | " | | | | | | | " | " | 1Y | " | 23 | " |
| | | " | 105 | " | | | | 2.7 V | | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 106 | " | | | | | | | " | OUT | | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 107 | " | | | | | | | " | | | | OUT | | 2.7 V | " | " | 4Y | " | " | " |
| | $t_{HZ}$ 5/ | " | 108 | 2.7 V | | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 11.5 | " |
| | | " | 109 | " | | | | | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 110 | " | | | | | | | " | OUT | GND | | | | | " | " | 3Y | " | " | " |
| | | " | 111 | " | | | | | | | " | | | | OUT | GND | | " | " | 4Y | " | " | " |
| | $t_{LZ}$ | " | 112 | GND | 2.7 V | | OUT | | | | " | | | | | | | " | " | 1Y | " | 17 | " |
| | | " | 113 | " | | | | 2.7 V | | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 114 | " | | | | | | | " | OUT | | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 115 | " | | | | | | | " | | | | OUT | | 2.7 V | " | " | 4Y | " | " | " |
| 10 | $t_{PHL1}$ | " | 116 | IN | GND | 2.7 V | OUT | | | | " | | | | | | | GND | " | 1Y | " | 18.5 | " |
| $T_C$ = +125°C | | " | 117 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 118 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 119 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PLH1}$ | " | 120 | " | GND | 2.7 V | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 121 | " | | | | GND | 2.7 V | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 122 | " | | | | | | | " | OUT | 2.7 V | GND | | | | " | " | 3Y | " | " | " |
| | | " | 123 | " | | | | | | | " | | | | OUT | 2.7 V | GND | " | " | 4Y | " | " | " |
| | $t_{PHL2}$ | " | 124 | GND | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 10 | " |
| | | " | 125 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 126 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 127 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{PLH2}$ | " | 128 | " | IN | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | " | " |
| | | " | 129 | " | | | | IN | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 130 | " | | | | | | | " | OUT | GND | IN | | | | " | " | 3Y | " | " | " |
| | | " | 131 | " | | | | | | | " | | | | OUT | GND | IN | " | " | 4Y | " | " | " |
| | $t_{ZH}$ | " | 132 | 2.7 V | | GND | OUT | | | | " | | | | | | | IN | " | 1Y | " | 28 | " |
| | | " | 133 | " | | | | | GND | | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 134 | " | | | | | | | " | OUT | GND | | | | | " | " | 3Y | " | " | " |
| | | " | 135 | " | | | | | | | " | | | | OUT | GND | | " | " | 4Y | " | " | " |

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TABLE III. Group A inspection for device type 07 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 7/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | S | 1A | 1B | 1Y | 2A | 2B | 2Y | GND | 3Y | 3B | 3A | 4Y | 4B | 4A | 0E | $V_{CC}$ | | | | |
| 10 | $t_{ZL}$ | 3003 | 136 | GND | 2.7 V | | OUT | | | | GND | | | | | | | IN | 5.0 V | 1Y | 2.0 | 30 | ns |
| $T_C$ = +125°C | | Fig. 10 | 137 | " | | | | 2.7 V | | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 138 | " | | | | | | | " | OUT | | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 139 | " | | | | | | | " | | | | OUT | | 2.7 V | " | " | 4Y | " | " | " |
| | $t_{HZ}$ 6/ | " | 140 | 2.7 V | | GND | OUT | | | | " | | | | | | | " | " | 1Y | " | 15 | " |
| | | " | 141 | " | | | | | GND | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 142 | " | | | | | | | " | OUT | GND | | | | | " | " | 3Y | " | " | " |
| | | " | 143 | " | | | | | | | " | | | | OUT | GND | | " | " | 4Y | " | " | " |
| | $t_{LZ}$ | " | 144 | GND | 2.7 V | | OUT | | | | " | | | | | | | " | " | 1Y | " | 22 | " |
| | | " | 145 | " | | | | 2.7 V | | OUT | " | | | | | | | " | " | 2Y | " | " | " |
| | | " | 146 | " | | | | | | | " | OUT | | 2.7 V | | | | " | " | 3Y | " | " | " |
| | | " | 147 | " | | | | | | | " | | | | OUT | | 2.7 V | " | " | 4Y | " | " | " |
| 11 | Same tests, terminal conditions, and limits as for subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

1/ For circuit B, $I_{OS(max)}$ = -110 mA.

2/ A = 2.4 V; B = 0.4 V.

3/ H ≥ 1.5 V; L ≤ 1.5 V.

4/ Only a summary of attributes is required.

5/ $t_{HZ}$ maximum limit for circuit C is 22 ns.

6/ $t_{HZ}$ maximum limit for circuit C is 24 ns.

7/ Case 2 pins not designated are NC.





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TABLE III. Group A inspection for device type 08
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | 1G | B | 1C3 | 1C2 | 1C1 | 1C0 | 1Y | GND | 2Y | 2C0 | 2C1 | 2C2 | 2C3 | A | 2G | $V_{CC}$ | | | | |
| 1<br>$T_C$ = +25°C | $V_{OH}$ | 3006 | 1 | 0.8 V | 0.8 V | | | | 2.0 V | -1 mA | GND | | | | | | 0.8 V | | 4.5 V | 1Y | 2.4 | | V |
| | | 3006 | 2 | | " | | | | | | " | -1 mA | 2.0 V | | | | " | 0.8 V | " | 2Y | 2.4 | | " |
| | $V_{OL}$ | 3007 | 3 | 0.8 V | " | | | | 0.8 V | 20 mA | " | | | | | | " | | " | 1Y | | 0.5 | " |
| | | 3007 | 4 | | " | | | | | | " | 20 mA | 0.8 V | | | | " | 0.8 V | " | 2Y | | 0.5 | " |
| | $I_{OFF1}$ | | 5 | 2.0 V | | | | | | 2.7 V | " | | | | | | | | 5.5 V | 1Y | | 50 | µA |
| | | | 6 | | | | | | | | " | 2.7 V | | | | | | 2.0 V | " | 2Y | | 50 | " |
| | $I_{OFF2}$ | | 7 | 2.0 V | | | | | | 0.5 V | " | | | | | | | | " | 1Y | | -50 | " |
| | | | 8 | | | | | | | | " | 0.5 V | | | | | | 2.0 V | " | 2Y | | -50 | " |
| | $V_{IC}$ | | 9 | | -18 mA | | | | | | " | | | | | | -18 mA | | 4.5 V | A | | -1.2 | V |
| | | | 10 | | | | | | | | " | | | | | | | | " | B | | " | " |
| | | | 11 | | | | | | -18 mA | | " | | | | | | | | " | 1C0 | | " | " |
| | | | 12 | | | | | -18 mA | | | " | | | | | | | | " | 1C1 | | " | " |
| | | | 13 | | | | -18 mA | | | | " | | | | | | | | " | 1C2 | | " | " |
| | | | 14 | | | -18 mA | | | | | " | | | | | | | | " | 1C3 | | " | " |
| | | | 15 | -18 mA | | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | | 16 | | | | | | | | " | | -18 mA | | | | | | " | 2C0 | | " | " |
| | | | 17 | | | | | | | | " | | | -18 mA | | | | | " | 2C1 | | " | " |
| | | | 18 | | | | | | | | " | | | | -18 mA | | | | " | 2C2 | | " | " |
| | | | 19 | | | | | | | | " | | | | | -18 mA | | | " | 2C3 | | " | " |
| | | | 20 | | | | | | | | " | | | | | | | -18 mA | " | 2G | | " | " |
| | $I_{IL}$ | 3009 | 21 | | | | | | | | " | | | | | | 0.5 V | | 5.5 V | A | -1.0 | -2.0 | mA |
| | | " | 22 | | 0.5 V | | | | | | " | | | | | | | | " | B | " | " | " |
| | | " | 23 | 0.5 V | | | | | | | " | | | | | | | | " | 1G | " | " | " |
| | | " | 24 | | | | | | | | " | | | | | | | 0.5 V | " | 2G | " | " | " |
| | | " | 25 | GND | GND | | | | 0.5 V | | " | | | | | | GND | | " | 1C0 | " | " | " |
| | | " | 26 | " | GND | | | 0.5 V | | | " | | | | | | 5.5 V | | " | 1C1 | " | " | " |
| | | " | 27 | " | 5.5 V | | 0.5 V | | | | " | | | | | | GND | | " | 1C2 | " | " | " |
| | | " | 28 | " | 5.5 V | 0.5 V | | | | | " | | | | | | 5.5 V | | " | 1C3 | " | " | " |
| | | " | 29 | | GND | | | | | | " | | 0.5 V | | | | GND | GND | " | 2C0 | " | " | " |
| | | " | 30 | | GND | | | | | | " | | | 0.5 V | | | 5.5 V | " | " | 2C1 | " | " | " |
| | | " | 31 | | 5.5 V | | | | | | " | | | | 0.5 V | | GND | " | " | 2C2 | " | " | " |
| | | " | 32 | | 5.5 V | | | | | | " | | | | | 0.5 V | 5.5 V | " | " | 2C3 | " | " | " |
| | $I_{IH1}$ | 3010 | 33 | | | | | | | | " | | | | | | 2.7 V | | " | A | | 50 | µA |
| | | " | 34 | | 2.7 V | | | | | | " | | | | | | | | " | B | | " | " |
| | | " | 35 | 2.7 V | | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | " | 36 | | | | | | | | " | | | | | | | 2.7 V | " | 2G | | " | " |
| | | " | 37 | 5.5 V | 5.5 V | | | | 2.7 V | | " | | | | | | 5.5 V | | " | 1C0 | | " | " |
| | | " | 38 | " | 5.5 V | | | 2.7 V | | | " | | | | | | GND | | " | 1C1 | | " | " |
| | | " | 39 | " | GND | | 2.7 V | | | | " | | | | | | 5.5 V | | " | 1C2 | | " | " |
| | | " | 40 | " | GND | 2.7 V | | | | | " | | | | | | GND | | " | 1C3 | | " | " |
| | | " | 41 | | 5.5 V | | | | | | " | | 2.7 V | | | | 5.5 V | 5.5 V | " | 2C0 | | " | " |
| | | " | 42 | | 5.5 V | | | | | | " | | | 2.7 V | | | GND | " | " | 2C1 | | " | " |
| | | " | 43 | | GND | | | | | | " | | | | 2.7 V | | 5.5 V | " | " | 2C2 | | " | " |
| | | " | 44 | | GND | | | | | | " | | | | | 2.7 V | GND | " | " | 2C3 | | " | " |

See footnotes at end of device type 08



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TABLE III. Group A inspection for device type 08 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | 1G | B | 1C3 | 1C2 | 1C1 | 1C0 | 1Y | GND | 2Y | 2C0 | 2C1 | 2C2 | 2C3 | A | 2G | $V_{CC}$ | | | | |
| 1 | $I_{IH2}$ | 3010 | 45 | | | | | | | | GND | | | | | | 5.5 V | | 5.5 V | A | | 1.0 | mA |
| $T_C$ = +25°C | | " | 46 | | 5.5 V | | | | | | " | | | | | | | | " | B | | " | " |
| | | " | 47 | 5.5 V | | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | " | 48 | | | | | | | | " | | | | | | | 5.5 V | " | 2G | | " | " |
| | | " | 49 | 5.5 V | 5.5 V | | | | 5.5 V | | " | | | | | | 5.5 V | | " | 1C0 | | " | " |
| | | " | 50 | " | 5.5 V | | | 5.5 V | | | " | | | | | | GND | | " | 1C1 | | " | " |
| | | " | 51 | " | GND | | 5.5 V | | | | " | | | | | | 5.5 V | | " | 1C2 | | " | " |
| | | " | 52 | " | GND | 5.5 V | | | | | " | | | | | | GND | | " | 1C3 | | " | " |
| | | | 53 | | 5.5 V | | | | | | " | | 5.5 V | | | | 5.5 V | 5.5 V | " | 2C0 | | " | " |
| | $I_{IH2}$ | " | 54 | | 5.5 V | | | | | | " | | | 5.5 V | | | GND | " | " | 2C1 | | 1.0 | mA |
| | | " | 55 | | GND | | | | | | " | | | | 5.5 V | | 5.5 V | " | " | 2C2 | | " | " |
| | | " | 56 | | " | | | | | | " | | | | | 5.5 V | GND | " | " | 2C3 | | " | " |
| | $I_{OS}$ | 3011 | 57 | GND | " | | | | 5.5 V | GND | " | | | | | | " | GND | " | 1Y | -40 | -100 4/ | " |
| | | " | 58 | GND | " | | | | | | " | GND | 5.5 V | | | | " | GND | " | 2Y | -40 | -100 4/ | " |
| | $I_{CC0}$ | 3005 | 59 | 5.5 V | " | GND | GND | GND | GND | | " | | GND | GND | GND | GND | " | 5.5 V | " | $V_{CC}$ | | 100 | " |
| | $I_{CC1}$ | 3005 | 60 | GND | " | GND | GND | GND | GND | | " | | GND | GND | GND | GND | " | GND | " | $V_{CC}$ | | 80 | " |
| | $I_{CEX}$ | | 61 | GND | " | | | | 5.5 V | 5.5 V | " | | | | | | " | | " | 1Y | | 250 | µA |
| | | | 62 | | " | | | | | | " | 5.5 V | 5.5 V | | | | " | GND | " | 2Y | | 250 | " |
| 2 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. $V_{IL}$ = 0.7 V, $V_{OL(max)}$ = 0.45 V. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 7 3/ | Truth | 3014 | 63 | B 1/ | B 1/ | A 1/ | A 1/ | A 1/ | B 1/ | L | GND | L | B 1/ | A 1/ | A 1/ | A 1/ | B | B 1/ | 4.5 V | See 2/ | | | |
| $T_C$ = +25°C | table | " | 64 | " | " | B | B | B | A | H | " | H | A | B | B | B | B | " | " | " | | | |
| | test | " | 65 | " | " | A | A | B | " | L | " | L | A | B | A | A | A | " | " | " | | | |
| | | " | 66 | " | " | B | B | A | " | H | " | H | B | A | B | B | A | " | " | " | | | |
| | | " | 67 | " | A | A | B | A | " | L | " | L | A | A | B | A | B | " | " | " | | | |
| | | " | 68 | " | " | B | A | B | B | H | " | H | B | B | A | B | B | " | " | " | | | |
| | | " | 69 | " | " | B | A | A | A | L | " | L | A | A | A | B | A | " | " | " | | | |
| | | " | 70 | " | " | A | B | B | B | H | " | H | B | B | B | A | A | " | " | " | | | |
| 8 | Same tests, terminal conditions, and limits as for subgroup 7, except $T_C$ = +125°C and $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

See footnotes at end of device type 08



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TABLE III. Group A inspection for device type 08 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | 1G | B | 1C3 | 1C2 | 1C1 | 1C0 | 1Y | GND | 2Y | 2C0 | 2C1 | 2C2 | 2C3 | A | 2G | $V_{CC}$ | | | | |
| 9 | $t_{PHL1}$ | 3003 | 71 | GND | GND | | | | IN | OUT | GND | | | | | | GND | | 5.0 V | 1Y | 2.0 | 11.0 | ns |
| $T_C$ = +25°C | | Fig. 10 | 72 | " | GND | | | IN | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 73 | " | 2.7 V | | IN | | | " | " | | | | | | GND | | " | " | " | " | " |
| | | " | 74 | " | 2.7 V | IN | | | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 75 | | GND | | | | | | " | OUT | IN | | | | GND | GND | " | 2Y | " | " | " |
| | | " | 76 | | GND | | | | | | " | " | | IN | | | 2.7 V | " | " | " | " | " | " |
| | | " | 77 | | 2.7 V | | | | | | " | " | | | IN | | GND | " | " | " | " | " | " |
| | | " | 78 | | 2.7 V | | | | | | " | " | | | | IN | 2.7 V | " | " | " | " | " | " |
| | $t_{PLH1}$ | " | 79 | GND | GND | | | | IN | OUT | " | | | | | | GND | | " | 1Y | " | " | " |
| | | " | 80 | " | GND | | | IN | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 81 | " | 2.7 V | | IN | | | " | " | | | | | | GND | | " | " | " | " | " |
| | | " | 82 | " | 2.7 V | IN | | | | " | " | | | | | | 2.7 V | | " | " | " | " | " |
| | | " | 83 | | GND | | | | | | " | OUT | IN | | | | GND | GND | " | 2Y | " | " | " |
| | | " | 84 | | GND | | | | | | " | " | | IN | | | 2.7 V | " | " | " | " | " | " |
| | | " | 85 | | 2.7 V | | | | | | " | " | | | IN | | GND | " | " | " | " | " | " |
| | | " | 86 | | 2.7 V | | | | | | " | " | | | | IN | 2.7 V | " | " | " | " | " | " |
| | $t_{PHL2}$ | " | 87 | GND | GND | | | GND | 2.7 V | OUT | " | | | | | | IN | | " | 1Y | " | 20 | " |
| | | " | 88 | GND | IN | | GND | | 2.7 V | OUT | " | | | | | | GND | | " | 1Y | " | " | " |
| | | " | 89 | | GND | | | | | | " | OUT | 2.7 V | GND | | | IN | GND | " | 2Y | " | " | " |
| | | " | 90 | | IN | | | | | | " | OUT | 2.7 V | | GND | | GND | GND | " | 2Y | " | " | " |

See footnotes at end of device type 08





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TABLE III. Group A inspection for device type 08 – Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | X, 2 5/ | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | Min | Max | |
| | | | Test no. | 1G | B | 1C3 | 1C2 | 1C1 | 1C0 | 1Y | GND | 2Y | 2C0 | 2C1 | 2C2 | 2C3 | A | 2G | $V_{CC}$ | | | | |
| 9 | $t_{PLH2}$ | 3003 | 91 | GND | GND | | | GND | 2.7 V | OUT | GND | | | | | | IN | | 5.0 V | 1Y | 2.0 | 20 | ns |
| $T_C$ = +25°C | | Fig. 10 | 92 | GND | IN | | GND | | 2.7 V | OUT | " | | | | | | GND | | " | 1Y | " | " | " |
| | | " | 93 | | GND | | | | | | " | OUT | 2.7 V | GND | | | IN | GND | " | 2Y | " | " | " |
| | | " | 94 | | IN | | | | | | " | OUT | 2.7 V | | GND | | GND | GND | " | 2Y | " | " | " |
| | $t_{ZH}$ | " | 95 | IN | GND | | | | 2.7 V | OUT | " | | | | | | " | | " | 1Y | " | 23 | " |
| | | " | 96 | " | " | | | | | | " | OUT | 2.7 V | | | | " | IN | " | 2Y | " | 23 | " |
| | $t_{ZL}$ | " | 97 | " | " | | | | GND | OUT | " | | | | | | " | | " | 1Y | " | 24 | " |
| | | " | 98 | " | " | | | | | | " | OUT | GND | | | | " | IN | " | 2Y | " | 24 | " |
| | $t_{HZ}$ | " | 99 | " | " | | | | 2.7 V | OUT | " | | | | | | " | | " | 1Y | " | 13.5 | " |
| | | " | 100 | " | " | | | | | | " | OUT | 2.7 V | | | | " | IN | " | 2Y | " | 13.5 | " |
| | $t_{LZ}$ | " | 101 | " | " | | | | GND | OUT | " | | | | | | " | | " | 1Y | " | 15.5 | " |
| | | " | 102 | " | " | | | | | | " | OUT | GND | | | | " | IN | " | 2Y | " | 15.5 | " |
| 10 $T_C$ = +125°C | Same tests and terminal conditions as subgroup 9, with limits as follows: $t_{PHL1}$ = 14.5 ns, $t_{PLH1}$ = 14.5 ns, $t_{PHL2}$ = 26 ns, $t_{PLH2}$ = 26 ns, $t_{ZH}$ = 30 ns, $t_{ZL}$ = 31 ns, $t_{HZ}$ = 18 ns, $t_{LZ}$ = 20 ns. | | | | | | | | | | | | | | | | | | | | | | |
| 11 | Same tests, terminal conditions, and limits as for subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

1/ A = 2.4 V; B = 0.4 V.

2/ H ≥ 1.5 V; L ≤ 1.5 V.

3/ Only a summary of attributes is required.

4/ For circuit B, IOS(max) = -110 mA.

5/ Case 2 pins not designated are NC.



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5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but it is not mandatory)

6.1 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. PIN and compliance identifier, if applicable (see 1.2).
c. Requirements for delivery of one copy of the conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
d. Requirement for certificate of compliance, if applicable.
e. Requirements for notification of change of product or process to acquiring activity in addition to notification to the qualifying activity, if applicable.
f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.
g. Requirements for product assurance options.
h. Requirements for special carriers, lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.
i. Requirements for "JAN" marking.
j. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43123-1199.

6.4 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.



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6.5 Abbreviations, symbols and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535 and MIL-HDBK-1331, and as follows:

GND ..... Electrical ground (common terminal)
$I_{IN}$ ..... Current flowing into an input terminal.
$V_{IN}$ ..... Voltage level at an input terminal.
$t_{ZH}$ ..... Output enable time (of a three-state output) to high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined high level.
$t_{ZL}$ ..... Output enable time (of a three-state output) to low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined low level.
$t_{HZ}$ ..... Output disable time (of a three-state output) from high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high level impedance (off) state.
$t_{LZ}$ ..... Output disable time (of a three-state output) from low level. The time between the specified reference points on the input and output voltage waveforms with the three state output changing from the defined low level to a high impedance (off) state.

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming shall not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic industry type. Generic industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

| Device type | Commercial type |
|---|---|
| 01 | 54S151 |
| 02 | 54S153 |
| 03 | 54S157 |
| 04 | 54S158 |
| 05 | 54S251 |
| 06 | 54S257 |
| 07 | 54S258 |
| 08 | 54S253 |

6.8 Manufacturers' designations. Manufacturers' circuits included in this specification are designated as shown in table IV.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.



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TABLE IV. Manufacturers' designations.

| Device type | Commercial Type | Circuits | | | | |
|---|---|---|---|---|---|---|
| | | A | B | C | D | E |
| | | Texas Instruments | Signetics Corp. | Advanced Micro Devices Inc. | Fairchild Semiconductor | National Semiconductor |
| 01 | 54S151 | X | X | X | X | X |
| 02 | 54S153 | X | X | X | X | X |
| 03 | 54S157 | X | X | X | X | X |
| 04 | 54S158 | X | X | X | X | X |
| 05 | 54S251 | X | X | X | X | |
| 06 | 54S257 | X | X | X | X | |
| 07 | 54S258 | X | X | X | X | |
| 08 | 54S253 | | X | X | X | |



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Custodians:
Army – CR
Navy - EC
Air Force – 11
DLA-CC

Preparing activity:
DLA - CC

(Project 5962-2005-007)

Review activities:
Army – MI, SM
Navy – AS, CG, MC, SH, TD
Air Force – 03, 19, 99

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .



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